

Qubits Are Not Enough: Deflation and Measurement Limits in Variational Quantum Modal Analysis

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ABSTRACT Amplitude encoding stores an N -degree-of-freedom structural mode using $\lceil \log_2 N \rceil$ qubits, but compact encoding does not by itself make quantum modal analysis reliable or scalable. This article develops a rigorous evaluation framework for variational quantum eigensolvers applied to finite-element structural dynamics. A Cholesky transformation maps the generalized eigenproblem to a symmetric mass-orthonormal Hamiltonian, so Euclidean quantum-state orthogonality is identical to structural mass orthogonality. A two-scale normalization separates optimizer conditioning from spectrally safe padding. For higher modes, the known deflation penalty condition is sharpened to a necessary-and-sufficient threshold, and a Davis–Kahan bound quantifies the effect of approximate lower-mode projectors. Numerical studies separate mapping correctness, ansatz expressibility, preprocessing, Pauli measurement growth, finite-shot frequency uncertainty, and damage localization. For structural chains with up to 128 degrees of freedom, consistent-mass whitening increases transformed-operator density from 0.023 to 0.308 and nonzero Pauli terms from 576 to 2950; a two- and three-dimensional generalization shows this densification worsening, and appearing later in N , as spatial dimension increases, and validates a subspace diagnostic against a genuine degenerate mode pair unreachable in one dimension. Exact state-vector experiments show that qubit count and circuit expressibility are distinct: shallow circuits fail even when the state dimension fits. At 10^5 shots per Pauli term, the first-mode frequency root-mean-square error remains 2.45% and a 10% stiffness loss is localized in 84.5% of trials, but certified localization on a six-DOF benchmark needs $\approx 10^8$ circuit executions. The results sharpen the deflation condition, give validation criteria, quantify resource bottlenecks, and claim no quantum speedup.

INDEX TERMS Finite-element eigenproblems, measurement complexity, structural dynamics, variational quantum eigensolvers.

I. INTRODUCTION

MODAL analysis underlies seismic response, wind and pedestrian serviceability, reduced-order simulation, operational modal analysis, finite-element model updating, and vibration-based structural health monitoring [1]–[4]. For a structure with N unconstrained degrees of freedom (DOFs), the low-frequency modes solve

$$\mathbf{K}\phi_i = \lambda_i \mathbf{M}\phi_i, \quad \lambda_i = \omega_i^2, \quad f_i = \frac{\omega_i}{2\pi}. \quad (1)$$

Classical Lanczos, subspace-iteration, and contour-based eigensolvers are mature and highly effective. A quantum method must therefore be evaluated against physical residuals, modal orthogonality, measurement cost, state-preparation cost, and end-to-end preprocessing, not against matrix dimension alone.

The variational quantum eigensolver (VQE) minimizes an expectation-valued Rayleigh quotient over a parameterized quantum state [5], [6], and is among the most studied near-term algorithms [7]–[9]. Excited-state extensions include variational quantum deflation (VQD) [10] and subspace-search VQE [11]. Generalized eigenvalue formulations have been developed for finite-element problems [12], and structural demonstrations have coupled finite-element software with VQE on simulators and small processors [13]. More recently, Taliadouros et al. used SSVQE and VQD inside Monte Carlo simulation to estimate multiple eigenvalue statistics of stochastic structural systems, including a proof-of-concept execution on IBM Sherbrooke [14]. Accordingly, neither the classical Cholesky reduction nor the use of VQD/SSVQE for multiple structural eigenvalues is claimed as new here. Higgott et al.

already introduce the deflated cost equation (21) and observe that $\beta_j > \varepsilon_r - \varepsilon_j$ suffices to expose the target mode [10]; that sufficiency condition is theirs, not ours. The contribution relative to this literature is the converse, showing the condition is also necessary and extending it to the padded spectrum; the deterministic certified penalty rule; the identification of the penalty that minimizes the approximate-projector certificate; the separation of padding certification from optimizer scaling; and end-to-end accounting of whitening, Pauli, finite-shot, and damage-decision costs. Existing studies establish feasibility, but do not by themselves establish scalable structural utility.

The central premise of this paper is summarized by its title: logarithmic qubit count is not sufficient. A usable structural modal solver must address six distinct questions:

- (i) Is the generalized eigenproblem mapped to a Hermitian operator without changing the physical mass metric?
- (ii) Are padded and previously extracted states excluded with provable, rather than heuristic, penalty conditions?
- (iii) Does whitening preserve or destroy sparsity and measurement economy?
- (iv) Can the ansatz represent the requested eigenspace and can the cost be trained?
- (v) Can modal observables be estimated at finite-shot precision relevant to engineering decisions?
- (vi) Can repeated monitoring distinguish genuine structural change from numerical mode swapping without biasing the eigensolution?

The contributions below are stated as demonstrated results. Each is either proved or measured in this paper; where a component is proposed but not evaluated, it is named separately afterwards rather than counted here.

- (i) a mass-orthonormal Cholesky mapping, with explicit accounting of its classical cost and of the sparsity it destroys;
- (ii) a two-scale spectral normalization that decouples the optimizer scale from certified safe padding;
- (iii) for higher modes, the converse to the known variational-deflation threshold—showing the condition is necessary as well as sufficient—its extension to the padded spectrum, a deterministic certified penalty rule, and the identification of the penalty β_j^* that minimizes the approximate-projector certificate, which is not the smallest admissible one;
- (iv) an exact closed form for the nonzero Pauli count of a lumped-mass mesh, $(n_q + 2)2^{n_q-1}$, with a checkable necessary-and-sufficient condition for it to be attained, showing that the quadratic measurement bound is reached only through the mass model;
- (v) end-to-end resource accounting—whitening, Pauli growth, commuting groups, state preparation, finite-shot frequency uncertainty, and damage-decision cost—extending beyond a single three-qubit demonstration;
- (vi) evidence that expressibility and trainability are in tension for this ansatz family: gradient concentration reaches the barren-plateau rate at depths still far below the depth

expressibility requires;

- (vii) a two- and three-dimensional generalization showing that densification worsens, and appears later in N , with spatial dimension, together with a degeneracy-diagnostic validation unreachable in one dimension.

Two further procedures are developed but deliberately not claimed above, because neither is evaluated experimentally: the lexicographic refinement of Section V, which replaces weighted sums of energy, variance and leakage by a constrained problem, and the post-hoc temporal mode-tracking rule of Section VII, which assigns mode labels after independent eigensolution instead of through continuity penalties inside it. They are included because the evaluation framework is incomplete without them, not as validated results.

Fig. 1 summarizes the end-to-end pipeline analyzed in the rest of the paper.

II. STRUCTURAL-DYNAMICS FOUNDATION

For a linear time-invariant structure,

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{C}\dot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{B}_f \mathbf{f}(t). \quad (2)$$

Under uniform support acceleration $\ddot{u}_g(t)$,

$$\mathbf{M}\ddot{\mathbf{u}} + \mathbf{C}\dot{\mathbf{u}} + \mathbf{K}\mathbf{u} = -\mathbf{M}\mathbf{r}\ddot{u}_g(t). \quad (3)$$

For undamped free vibration, substitute $\mathbf{u}(t) = \phi e^{i\omega t}$ into equation (2) to obtain equation (1). For a stable restrained structure, $\mathbf{M}$ and $\mathbf{K}$ are symmetric, $\mathbf{M} \succ 0$, and the modes may be mass normalized:

$$\Phi_m^T \mathbf{M} \Phi_m = \mathbf{I}_m, \quad (4)$$

$$\Phi_m^T \mathbf{K} \Phi_m = \Omega_m^2, \quad \Omega_m = \text{diag}(\omega_1, \dots, \omega_m). \quad (5)$$

The modal transformation $\mathbf{u} = \Phi_m \mathbf{q}$ yields, under classical damping,

$$\ddot{\mathbf{q}} + 2\Xi\Omega_m\dot{\mathbf{q}} + \Omega_m^2\mathbf{q} = \Phi_m^T \mathbf{B}_f \mathbf{f}, \quad (6)$$

where $\Xi = \text{diag}(\xi_1, \dots, \xi_m)$.

Finite-element assembly gives

$$\mathbf{K} = \sum_{e=1}^{n_e} \mathbf{A}_e^T \mathbf{K}_e \mathbf{A}_e, \quad \mathbf{M} = \sum_{e=1}^{n_e} \mathbf{A}_e^T \mathbf{M}_e \mathbf{A}_e. \quad (7)$$

Boundary conditions are imposed before the quantum mapping. Penalty-enforced supports are avoided because artificial high stiffness can worsen conditioning and contaminate the spectrum.

III. MASS-ORTHONORMAL HERMITIAN MAPPING AND PREPROCESSING

A. CHOLESKY MAPPING

The nonsymmetric product $\mathbf{M}^{-1}\mathbf{K}$ has the correct eigenvalues but is generally not Hermitian in the Euclidean metric. Let

$$\mathbf{M} = \mathbf{L}\mathbf{L}^T \quad (8)$$

be a Cholesky factorization and define

$$\mathbf{y}_i = \mathbf{L}^T \phi_i, \quad \mathbf{A} = \mathbf{L}^{-1} \mathbf{K} \mathbf{L}^{-T}. \quad (9)$$

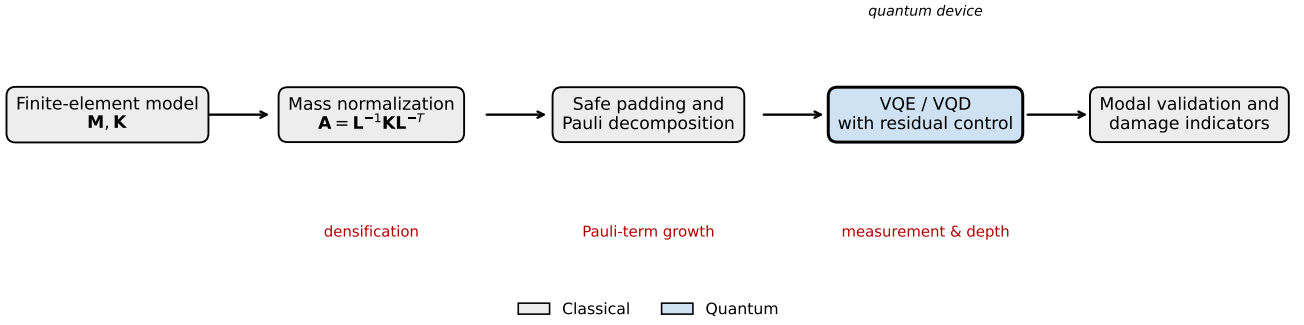


FIGURE 1. End-to-end pipeline. A finite-element model $(\mathbf{M}, \mathbf{K})$ is mass-normalized by the Cholesky map $\mathbf{A} = \mathbf{L}^{-1} \mathbf{K} \mathbf{L}^{-T}$, spectrally padded and Pauli-decomposed, solved by a variational eigensolver (VQE/VQD) with residual control, and finally converted into modal-validation and damage indicators. Each stage is analyzed for correctness, cost, and measurement precision separately. Red labels mark the dominant bottleneck stages; their asymptotic costs are quantified in Table 1.

Then

$$\mathbf{A} \mathbf{y}_i = \lambda_i \mathbf{y}_i. \quad (10)$$

Proposition 1 (Mass–Euclidean isometry). *If $\mathbf{M} \succ 0$ and $\mathbf{K} = \mathbf{K}^T$, then $\mathbf{A} = \mathbf{A}^T$ and*

$$\mathbf{y}_i^T \mathbf{y}_j = \phi_i^T \mathbf{M} \phi_j. \quad (11)$$

Thus Euclidean orthogonality of quantum states is exactly structural mass orthogonality after back-transformation.

Proof. Symmetry follows from $\mathbf{A}^T = \mathbf{L}^{-1} \mathbf{K}^T \mathbf{L}^{-T} = \mathbf{A}$. Moreover, $\mathbf{y}_i^T \mathbf{y}_j = \phi_i^T \mathbf{L} \mathbf{L}^T \phi_j = \phi_i^T \mathbf{M} \phi_j$. $\square$

The transformation should be implemented with triangular solves, not explicit inverses. Dense Cholesky has $O(N^3)$ arithmetic cost and $O(N^2)$ storage. Sparse cost depends on mesh dimension, ordering, and fill. For a class of two-dimensional finite-element meshes under nested dissection, Cholesky requires $O(N^{3/2})$ operations and $O(N \log N)$ nonzeros [15]. These costs are classical preprocessing and must be included in any end-to-end comparison.

B. WHITENING CAN DESTROY LOCALITY

Sparsity of $\mathbf{M}$ and $\mathbf{K}$ does not imply sparsity of $\mathbf{A}$. When $\mathbf{M}$ is diagonal, whitening is a diagonal scaling and often preserves graph locality. With a consistent mass matrix, $\mathbf{L}^{-1}$ can introduce long-range couplings. Define

$$\delta_A(\tau) = \frac{1}{N^2} \#\{(i, j) : |A_{ij}| > \tau \|\mathbf{A}\|_{\max}\}. \quad (12)$$

The transformed density δ_A is therefore an algorithmic observable, not a minor implementation detail. Every density reported in this paper uses $\tau = 10^{-12}$.

A generalized Rayleigh quotient can avoid explicit whitening,

$$\mathcal{R}_G(\theta) = \frac{\langle \psi(\theta) | \mathbf{K} | \psi(\theta) \rangle}{\langle \psi(\theta) | \mathbf{M} | \psi(\theta) \rangle}, \quad (13)$$

as in generalized variational formulations [12]. The trade-off is that both operators must be measured, the denominator must remain bounded away from zero, and higher-mode

orthogonality must be enforced in the $\mathbf{M}$ metric. The Cholesky and generalized-quotient routes should therefore be compared rather than assuming that whitening is free.

IV. TWO-SCALE SPECTRAL NORMALIZATION AND SAFE PADDING

A. TWO-SCALE NORMALIZATION

A Gershgorin upper bound is

$$U_G = \max_i \left(A_{ii} + \sum_{j \neq i} |A_{ij}| \right) \geq \lambda_{\max}(\mathbf{A}). \quad (14)$$

Using U_G as the sole scaling factor is safe but may compress the physical spectrum near zero. We therefore distinguish:

- an optimization scale $s_t > 0$, estimated for example by a short power or Lanczos iteration; and
- the certified physical bound $U = U_G/s_t$.

The optimization Hamiltonian is

$$\tilde{\mathbf{A}} = \frac{\mathbf{A}}{s_t}. \quad (15)$$

No assumption that $s_t \geq \lambda_{\max}$ is required.

B. SPECTRAL-SAFE PADDING

Let $n_q = \lceil \log_2 N \rceil$ and $D = 2^{n_q}$. Define

$$\mathbf{H}_p = \begin{bmatrix} \tilde{\mathbf{A}} & \mathbf{0} \\ \mathbf{0} & \alpha \mathbf{I}_{D-N} \end{bmatrix}, \quad \alpha = U + \delta_p, \quad \delta_p > 0. \quad (16)$$

Proposition 2 (Safe padding under arbitrary positive scaling). *Every physical eigenvalue of $\tilde{\mathbf{A}}$ is at most U , whereas every padded eigenvalue equals $U + \delta_p$. Hence padding cannot enter the physical low spectrum even when s_t underestimates $\lambda_{\max}(\mathbf{A})$.*

C. EFFECT OF CONSERVATIVE SCALING

Proposition 3 (Scalar-scaling law). *Let $\mathbf{H}_c = \mathbf{H}/c$ with $c > 0$ and $E_c(\theta) = \langle \psi(\theta) | \mathbf{H}_c | \psi(\theta) \rangle$. Then eigenvectors and condition number are unchanged, while*

$$\lambda_i(\mathbf{H}_c) = \lambda_i(\mathbf{H})/c, \quad \nabla E_c = \nabla E/c, \quad \text{Var}_{\psi}(\mathbf{H}_c) = \text{Var}_{\psi}(\mathbf{H})/c^2. \quad (17)$$

Consequently, excessive scaling does not change the exact eigenvectors, but it reduces absolute gaps and gradients and can interact badly with fixed optimizer tolerances and additive calibration errors. Under ideal shot noise, both signal and standard deviation scale as $1/c$, so relative precision is unchanged if all tolerances and coefficients are rescaled coherently. In practice, coherent rescaling requires the energy stopping tolerance, deflation margins, padding offset, confidence intervals, coefficient-pruning threshold, and any gradient threshold to be transformed by the same factor. The invariance fails when software retains fixed absolute tolerances, when coefficient truncation is not rescaled, or when hardware contributes an additive bias or noise floor that does not scale with the Hamiltonian coefficients. The optimization concern is therefore numerical and hardware-relative, not a universal loss of statistical information.

The physical frequency is recovered from $\tilde{\lambda}_i$ by

$$f_i = \frac{1}{2\pi} \sqrt{s_t \tilde{\lambda}_i}. \quad (18)$$

V. EXACT DEFLATION CONDITIONS FOR MULTIPLE MODES

A. GROUND-STATE VQE

For a normalized parameterized state $|\psi(\theta)\rangle = U(\theta)|0\rangle^{\otimes n_q}$,

$$E(\theta) = \langle \psi(\theta) | \mathbf{H}_p | \psi(\theta) \rangle \quad (19)$$

upper-bounds the lowest eigenvalue accessible within the ansatz manifold. This is a variational statement, not a guarantee that a nonconvex classical optimizer finds the global minimum.

B. EXACT DEFLATION THEOREM

Let the non-padded eigenpairs of $\mathbf{H}_p$ be $(\varepsilon_k, |u_k\rangle)$ with

$$\varepsilon_1 \leq \varepsilon_2 \leq \dots \leq \varepsilon_N < \alpha. \quad (20)$$

For target mode r , define the exact deflated operator

$$\mathbf{H}_r = \mathbf{H}_p + \sum_{j=1}^{r-1} \beta_j P_j, \quad P_j = |u_j\rangle\langle u_j|. \quad (21)$$

Theorem 1 (Necessary and sufficient deflation threshold). *Assume $\varepsilon_r < \varepsilon_{r+1}$, with $\varepsilon_{N+1} := \alpha$ when $D > N$ and $\varepsilon_{N+1} := +\infty$ otherwise. Then $|u_r\rangle$ is the unique ground state of $\mathbf{H}_r$ if and only if*

$$\beta_j > \varepsilon_r - \varepsilon_j \quad \text{for all } j < r. \quad (22)$$

Proof. The eigenvectors of $\mathbf{H}_r$ remain $|u_k\rangle$. For $k < r$, the corresponding eigenvalue is $\varepsilon_k + \beta_k$; for $k \geq r$, it is ε_k ; padded

states have eigenvalue α . The target is the unique minimum precisely when every shifted lower eigenvalue exceeds ε_r and $\varepsilon_{r+1}, \alpha > \varepsilon_r$. $\square$

The sufficiency direction is due to Higgott et al. [10], who introduce the deflated cost equation (21) and note that $\beta_j > \varepsilon_r - \varepsilon_j$ suffices. What equation (22) adds is the converse—no smaller penalty works, so the condition is sharp—and the inclusion of the padded block through the convention $\varepsilon_{N+1} := \alpha$, which matters because padding contributes eigenvalues that deflation must also clear.

The theorem replaces an arbitrary common penalty with a spectral condition. If a certified target upper bound $U_r \geq \varepsilon_r$ and prior eigenvalue error bounds $|\hat{\varepsilon}_j - \varepsilon_j| \leq \delta_j$ are available, a deterministic sufficient choice is

$$\beta_j = U_r - \hat{\varepsilon}_j + \delta_j + \tau, \quad \tau > 0. \quad (23)$$

Indeed, $\varepsilon_j + \beta_j \geq U_r + \tau > \varepsilon_r$. Two bound policies are distinguished. A quantum-only spectral setup may use

$$U_r^{(G)} = U_G/s_t, \quad (24)$$

which requires no auxiliary classical eigensolve but can be conservative. Alternatively, if $\vartheta_r^{(p)}$ is the r th Ritz value in a p -dimensional trial subspace, the min-max principle gives $\varepsilon_r \leq \vartheta_r^{(p)}$; hence

$$U_r^{(R)} = \vartheta_r^{(p)} + \delta_{\text{fp}} \quad (25)$$

is a tighter certified bound after allowing for floating-point error. Tighter is not automatically better here: by equation (32) a tighter U_r combined with a small τ drives the deflated separation g_r toward zero and the Davis–Kahan certificate toward vacuity, whereas the conservative Gershgorin policy equation (24) overshoots the threshold and lands in the safe regime. A tight bound policy must therefore be paired with a τ of order G_r . A sparse p -step Lanczos/Ritz pass costs approximately $O(p \text{nnz}(\mathbf{A}) + Np^2 + p^3)$ arithmetic ($\text{nnz}(\cdot)$ denoting the number of nonzero entries) and $O(Np)$ storage, and is therefore recorded explicitly as C_{bound} in the resource vector rather than treated as free preprocessing.

For a previously extracted mode, a practical high-confidence error radius can combine an eigenstate residual and Pauli-sampling uncertainty:

$$\delta_j^{(\alpha_j)} = \hat{r}_j + \sqrt{2 \log(2/\alpha_j) \sum_{\ell} \frac{h_{\ell}^2}{n_{\ell}}}, \quad (26)$$

$$\hat{r}_j = \|(\mathbf{H}_p - \hat{\varepsilon}_j \mathbf{I})|\hat{u}_j\rangle\|_2.$$

The second term is a conservative Hoeffding radius for independent Pauli sampling. The residual locates the Rayleigh quotient near some eigenvalue; using it as a bound for mode j additionally requires the estimated state to lie in the isolated modal interval assigned to j . In the exact threshold experiment below, exact eigenvalues are used only to verify theorem sharpness, so $\delta_j = 0$; operational studies must report whether equation (24) or equation (25) is used and include the corresponding classical cost.

C. APPROXIMATE PROJECTORS AND THE COST OF OVERSIZED PENALTIES

In practice, previous states are approximate. Let $\hat{P}_j$ be the estimated projectors and

$$\hat{\mathbf{H}}_r = \mathbf{H}_p + \sum_{j < r} \beta_j \hat{P}_j, \quad \Delta_r = \hat{\mathbf{H}}_r - \mathbf{H}_r. \quad (27)$$

Then

$$\|\Delta_r\|_2 \leq \sum_{j < r} \beta_j \|\hat{P}_j - P_j\|_2. \quad (28)$$

Define the exact deflated separation

$$g_r = \min \left\{ \min_{j < r} (\varepsilon_j + \beta_j - \varepsilon_r), \varepsilon_{r+1} - \varepsilon_r, \alpha - \varepsilon_r \right\}. \quad (29)$$

If $\|\Delta_r\|_2 < g_r/2$, Weyl's inequality preserves an isolated target eigenvalue and the Davis–Kahan theorem gives the safe bound [16]

$$\sin \angle(\hat{u}_r, u_r) \leq \frac{2 \|\Delta_r\|_2}{g_r}. \quad (30)$$

Equation (28) is easily misread as an argument for the smallest admissible penalty, since $\|\Delta_r\|_2$ grows linearly with β_j . That reading is incomplete, because β_j also enters the denominator of equation (30) through g_r . Write

$$G_r = \min(\varepsilon_{r+1} - \varepsilon_r, \alpha - \varepsilon_r) \quad (31)$$

for the separation available from the physical spectrum. Then g_r increases with β_j until it saturates at G_r , so the certified bound is

$$\text{minimized at } \beta_j^* = (\varepsilon_r - \varepsilon_j) + G_r, \quad (32)$$

diverges as β_j approaches the threshold from above, and grows only linearly beyond β_j^* . The smallest certified penalty is therefore the worst admissible choice, not the best: each penalty should be raised until the deflated separation saturates the physical gap, and not much further. In equation (23) this corresponds to sizing the total excess $U_r - \varepsilon_r + \delta_j + \tau$ against G_r , rather than driving τ to zero.

Two caveats attach to equation (32). It locates the minimum of the certificate, not of the error: in the perturbation experiment of Section X the measured target rotation continues to fall as β_j is increased past β_j^* , so equation (30) is conservative in a β -dependent way. And β_j^* requires ε_r and G_r , which are not available before the solve; the practical reading is that a conservative bound policy is safer than a tight one, not that the optimum can be hit exactly.

D. NO ARBITRARY VARIANCE OR LEAKAGE WEIGHTS

Hamiltonian variance remains a valuable residual:

$$\sigma_H^2(\psi) = \langle \mathbf{H}_p^2 \rangle - \langle \mathbf{H}_p \rangle^2 = \|\langle \mathbf{H}_p - E\mathbf{I} \rangle |\psi\rangle\|_2^2. \quad (33)$$

Padding leakage is

$$\ell_p(\psi) = \langle \psi | P_{\text{pad}} | \psi \rangle. \quad (34)$$

Variance has also been studied as an objective in its own right [17]; the difficulty here is not variance minimization but

the additive combination. Adding $\gamma\sigma_H^2 + \zeta\ell_p$ to the energy creates uncalibrated trade-offs and changes the optimization landscape. The primary algorithm therefore uses only

$$C_r(\theta) = \langle \psi(\theta) | \hat{\mathbf{H}}_r | \psi(\theta) \rangle. \quad (35)$$

An optional lexicographic refinement is

$$\begin{aligned} \min_{\theta} \sigma_H^2(\theta) \\ \text{s.t. } C_r(\theta) \leq C_r^* + \epsilon_E, \quad \ell_p(\theta) \leq \epsilon_p, \\ \max_{j < r} |\langle \hat{u}_j | \psi(\theta) \rangle|^2 \leq \epsilon_O. \end{aligned} \quad (36)$$

This preserves the energy ordering within a declared tolerance rather than hiding the trade-off in application-defined weights.

Remark 1 (Degenerate and near-degenerate targets). *Theorem 1 assumes $\varepsilon_r < \varepsilon_{r+1}$, and equation (30) requires $g_r > 0$. Neither holds for a degenerate target, and Section XI-C exhibits a genuine pair with relative gap 2.2×10^{-16} arising from nothing more exotic than a reflection symmetry. Symmetric geometries—square plates, circular tanks, regular framed buildings—make such pairs common rather than exceptional. For a repeated eigenvalue the projector P_j is not basis-independent within the shared eigenspace, so sequential deflation cannot isolate an individual member at all; only the invariant subspace is well defined. Block deflation, replacing $\beta_j P_j$ by $\beta_j P_{S_j}$ onto whole eigenspaces, is the natural generalization and is not developed here. This is the same failure mode that motivates the subspace diagnostic of equation (47).*

Remark 2 (Scope of the guarantee). *Theorem 1 identifies the correct ground state of the full deflated Hilbert-space operator. It does not prove global convergence of a finite-depth ansatz or a nonconvex classical optimizer. Ansatz approximation error and optimization failure are measured separately through restarts, residuals, and depth studies.*

E. SIMULTANEOUS AND TARGETED ALTERNATIVES

SSVQE maps orthogonal input states through a common unitary, preserving orthogonality by construction [11]. For a target shift σ , the folded operator

$$\mathbf{H}_\sigma = (\mathbf{H}_p - \sigma\mathbf{I})^2 \quad (37)$$

selects the mode nearest σ . These alternatives are useful when sequential projector error is unacceptable or when a narrow frequency band is known in advance.

VI. TRAINABILITY AND ANSATZ EXPRESSIBILITY

A real structural Hamiltonian permits a real-amplitude circuit,

$$U_L(\theta) = \left[\prod_{q=1}^{n_q} R_y(\theta_{qL}) \right] \prod_{\ell=1}^L \left[U_{\text{ent}}^{(\ell)} \prod_{q=1}^{n_q} R_y(\theta_{q,\ell-1}) \right]. \quad (38)$$

This is the real-amplitude member of the hardware-efficient family [18]. Logarithmic qubit count does not imply shallow representability. A normalized real vector in $D = 2^{n_q}$ dimensions has $D - 1$ independent amplitudes, whereas a

depth- L hardware-efficient circuit has only $O(Ln_q)$ parameters. Therefore, a fixed-depth family cannot be universally expressive as N grows.

Random parameterized circuits may exhibit barren plateaus in which gradient variance decays exponentially with qubit count [19]. Cost-function locality can materially alter trainability, and even shallow circuits with global costs may exhibit barren plateaus [20]. Binary amplitude encoding and Cholesky whitening can turn a local structural operator into a nonlocal qubit Hamiltonian, so structural locality does not guarantee a local quantum cost. This concern is directly connected to the scaling results in Section X: for consistent mass, the transformed density grows to 0.308 and the Pauli count to 2950 at $N = 128$, indicating that increasingly many computational-basis amplitudes and Pauli supports become coupled. Such densification moves the objective away from the local-cost setting in which trainability can be more favorable and toward the global-cost regime associated with gradient concentration. The present 2–12-qubit data cannot establish a structural barren plateau, but they show that whitening-induced nonlocality can worsen the locality condition as N grows independently of whether circuit depth is increased. Quantifying this interaction for larger qubit counts, alternative orderings, and domain-decomposed mappings is therefore an open problem rather than an assumed asymptotic advantage.

For a single parameter θ_a —here $a = 0$, the first R_y angle of the initial rotation layer—the empirical trainability statistic used here is

$$V_g(n_q) = \text{Var}_{\theta \sim \mathcal{D}} \left[\frac{\partial E(\theta)}{\partial \theta_a} \right]. \quad (39)$$

A finite $n_q \leq 12$ study cannot prove or disprove an asymptotic barren plateau, but it can detect whether an exponential decay has already set in. Over the tested range it has not (Section X): the uniform-initialization gradient variance is better described by a polynomial than by an exponential in n_q , and its high-qubit slope is far shallower than an exponential collapse.

VII. RECOVERY, STRUCTURAL IDENTIFICATION, AND HEALTH-MONITORING OBSERVABLES

A. PHYSICAL RECOVERY AND VALIDATION VECTOR

After removing padded coordinates, recover and mass-normalize

$$\phi_i = \mathbf{L}^{-T} \mathbf{y}_i, \quad \phi_i \leftarrow \frac{\phi_i}{\sqrt{\phi_i^T \mathbf{M} \phi_i}}. \quad (40)$$

The normalized equilibrium residual is

$$\rho_i = \frac{\left\| \mathbf{K} \phi_i^Q - \lambda_i^Q \mathbf{M} \phi_i^Q \right\|_2}{\left\| \mathbf{K} \phi_i^Q \right\|_2 + |\lambda_i^Q| \left\| \mathbf{M} \phi_i^Q \right\|_2}. \quad (41)$$

Multi-mode errors are

$$\epsilon_M = \left\| (\Phi^Q)^T \mathbf{M} \Phi^Q - \mathbf{I} \right\|_F, \quad (42)$$

$$\epsilon_K = \frac{\left\| (\Phi^Q)^T \mathbf{K} \Phi^Q - (\Omega^Q)^2 \right\|_F}{\left\| (\Omega^Q)^2 \right\|_F}. \quad (43)$$

No scalar exponential integrity index is introduced. Instead, every mode is reported through the non-aggregated diagnostic vector

$$\mathcal{D}_i = (\epsilon_{f,i}, \rho_i, \sigma_{H,i}^2, \ell_{p,i}, o_i, 1 - \text{MAC}_{ii}^{(M)}), \quad (44)$$

where $o_i = \max_{j < i} |\langle \hat{u}_j | \hat{u}_i \rangle|^2$. Acceptance thresholds are application-specific and must be stated component by component.

The mass-weighted modal assurance criterion is [21]

$$\text{MAC}_{ij}^{(M)} = \frac{|(\phi_i^C)^T \mathbf{M} \phi_j^Q|^2}{[(\phi_i^C)^T \mathbf{M} \phi_i^C][(\phi_j^Q)^T \mathbf{M} \phi_j^Q]}. \quad (45)$$

For a near-degenerate group, compare invariant subspaces. If

$$(\mathbf{Y}_C)^T \mathbf{Y}_Q = \mathbf{U} \cos \Theta \mathbf{V}^T, \quad (46)$$

then

$$\epsilon_{\text{sub}} = \|\sin \Theta\|_F \quad (47)$$

is invariant to rotations inside the repeated eigenspace.

B. PARTICIPATION AND FREQUENCY RESPONSE

For base excitation,

$$\Gamma_i = \frac{\phi_i^T \mathbf{M} \mathbf{r}}{\phi_i^T \mathbf{M} \phi_i}, \quad M_i^* = \Gamma_i^2 \quad (48)$$

for mass-normalized modes. The truncated modal frequency-response function is

$$\mathbf{H}_m(\omega) = \sum_{i=1}^m \frac{\phi_i \phi_i^T}{\omega_i^2 - \omega^2 + 2i\xi_i \omega_i \omega}. \quad (49)$$

Frequency accuracy without participation, shape, and response accuracy is not sufficient for structural use.

C. CONNECTION TO OUTPUT-ONLY IDENTIFICATION

For measured output power spectral density,

$$\mathbf{S}_{yy}(\omega) = \mathbf{U}(\omega) \mathbf{\Sigma}(\omega) \mathbf{U}^H(\omega), \quad (50)$$

and the dominant singular vector near resonance approximates the measured mode shape under the assumptions of frequency-domain decomposition [22]. Stochastic subspace identification uses

$$\mathbf{x}_{k+1} = \mathbf{A}_d \mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{y}_k = \mathbf{C}_d \mathbf{x}_k + \mathbf{v}_k, \quad (51)$$

with continuous poles $s_i = \log \mu_i / \Delta t$, frequencies $f_i = |s_i| / (2\pi)$, and damping ratios $\xi_i = -\Re(s_i) / |s_i|$ [23]–[25]. Experimental covariance estimates may define uncertainty-weighted MAC through $\mathbf{W}_i = \mathbf{\Sigma}_{\phi_i}^{-1}$ [26].

D. UNBIASED TEMPORAL MODE TRACKING

Every monitoring epoch is solved independently using the same independently defined objective. A previous solution may initialize the circuit, but it does not enter the objective. After solution, define for candidate pair (i, j)

$$\mathbf{z}_{ij,t} = \begin{bmatrix} \log(f_{j,t}/f_{i,t-1}) \\ 1 - \text{MAC}_{ij,t}^{(M)} \end{bmatrix}, \quad c_{ij,t} = \mathbf{z}_{ij,t}^T \Sigma_{i,0}^{-1} \mathbf{z}_{ij,t}, \quad (52)$$

where $\Sigma_{i,0}$ is estimated from healthy baseline variability. Cost functions combining a frequency ratio with a MAC term, minimized by an assignment algorithm, are standard in automated operational modal analysis [25]; what is adapted here is the uncertainty normalization and the insistence that pairing occur strictly after an independent eigensolve. Mode labels are assigned by

$$\pi_t^* = \arg \min_{\pi} \sum_i c_{i,\pi(i),t}. \quad (53)$$

A change alarm is based on $c_{i,\pi(i),t}$ exceeding a baseline quantile. A χ^2 calibration is not appropriate here: $1 - \text{MAC}$ is non-negative and concentrates at zero, so the quadratic form is not χ^2 distributed and the empirical quantile should be used instead. Because pairing occurs after independent eigensolution, the objective cannot pull a damaged mode toward its healthy predecessor. This is weaker than unbiasedness of the alarm: the assignment itself can still mislabel under damage-induced mode veering, after which the change statistic is referenced to the wrong baseline mode, and modes that go unidentified in an epoch—routine in operational modal analysis—need an explicit missing-label rule that equation (53) does not provide. A further practical restriction is that $\text{MAC}^{(M)}$ requires $\mathbf{M}$ at every degree of freedom, whereas output-only identification returns shapes only at instrumented coordinates; a sensor-space unweighted MAC must be substituted there, at the cost of the mass-orthogonality interpretation. Near degeneracy, the same procedure is applied to principal angles between subspaces.

E. SENSOR-SPACE RECOVERY

For sensor selector $\mathbf{c}_s$,

$$\phi_{i,s} = \mathbf{c}_s^T \phi_i = \mathbf{c}_s^T \mathbf{L}^{-T} \mathbf{y}_i = \mathbf{d}_s^T \mathbf{y}_i, \quad \mathbf{d}_s = \mathbf{L}^{-1} \mathbf{c}_s. \quad (54)$$

Therefore

$$\phi_{i,s} = \|\mathbf{d}_s\|_2 \langle \hat{\mathbf{d}}_s | \psi_i \rangle. \quad (55)$$

This can reduce readout when only instrumented coordinates are needed, although preparing $|\hat{\mathbf{d}}_s\rangle$ has its own cost. It is resource-favorable only when $\mathbf{d}_s = \mathbf{L}^{-1} \mathbf{c}_s$ has exploitable sparsity, low-rank structure, or a reusable preparation oracle; if whitening makes $\mathbf{d}_s$ dense and unstructured, generic amplitude preparation can require $O(N)$ gates and may erase the advantage over recovering the required classical amplitudes directly.

F. DAMAGE SENSITIVITY AND MODAL STRAIN ENERGY

Let element damage reduce stiffness:

$$\mathbf{K}(\mathbf{d}) = \sum_{e=1}^{n_e} (1 - d_e) \mathbf{A}_e^T \mathbf{K}_e^0 \mathbf{A}_e. \quad (56)$$

For mass-normalized modes and stiffness-only damage,

$$\frac{\partial \lambda_i}{\partial d_e} = -\phi_i^T \mathbf{K}_e^0 \phi_i. \quad (57)$$

The element modal-strain-energy fraction is formed with the baseline element stiffness $\mathbf{K}_e^0$ in the numerator, for both the healthy and the damaged structure,

$$\eta_{e,i} = \frac{\phi_i^T \mathbf{K}_e^0 \phi_i}{\phi_i^T \mathbf{K} \phi_i}, \quad (58)$$

in the spirit of standard modal-strain-energy practice [27], [28], though not identical to it: those formulations hold the undamaged stiffness fixed in both numerator and denominator and compare the resulting fractions as a ratio, whereas equation (58) retains the current global $\mathbf{K}$ in the denominator—where it is the measured eigenvalue—and equation (61) takes an absolute difference. The absolute difference is biased toward elements carrying high modal strain energy, which a ratio formulation would partly normalize away. This is not a convention of convenience: $\mathbf{K}_e(\mathbf{d})$ is the unknown being inferred, so an operator built from it could not be assembled, let alone measured. The denominator is unaffected, since $\phi_i^T \mathbf{K} \phi_i = \lambda_i$ for mass-normalized modes and is therefore supplied by the eigensolver itself. With

$$\mathbf{H}_e = \frac{\mathbf{L}^{-1} \mathbf{K}_e^0 \mathbf{L}^{-T}}{s_t}, \quad (59)$$

one obtains

$$\eta_{e,i}^{(\text{ep})} = \frac{\langle \psi_i^{(\text{ep})} | \mathbf{H}_e | \psi_i^{(\text{ep})} \rangle}{\langle \psi_i^{(\text{ep})} | \tilde{\mathbf{A}}^{(\text{ep})} | \psi_i^{(\text{ep})} \rangle}, \quad (60)$$

Two caveats bound the practical reach of equation (60). The numerator operator is built from the baseline model and is therefore available, but the state $|\psi_i\rangle$ is not. Preparing the damaged mode requires either an eigensolve of the damaged operator, which presumes exactly the model being inferred, or amplitude encoding of a mode shape identified classically from measured response—in which case $\phi_i^T \mathbf{K}_e^0 \phi_i$ is an $O(\text{nnz}(\mathbf{K}_e))$ classical inner product and no measurement strategy can make the quantum evaluation competitive. Operational modal analysis moreover returns shapes only at instrumented coordinates, so the full N -vector that amplitude encoding needs does not exist. The finite-shot study of Section X accordingly isolates measurement variance under a known damaged model; it quantifies the cost of the expectation-value formulation and is not a demonstration of a deployable monitoring method.

A multimode change score is

$$D_e = \sum_{i=1}^m w_i |\eta_{e,i}^d - \eta_{e,i}^0|, \quad \sum_i w_i = 1. \quad (61)$$

This class of energy-based localization is related to established modal strain-energy methods [27], [28]. The finite-shot study below tests whether the quantum expectation formulation remains usable when the exact modes are no longer assumed available.

VIII. MEASUREMENT AND RESOURCE ACCOUNTING

A. PAULI REPRESENTATION

For n_q qubits,

$$\mathbf{H} = \sum_{\ell=1}^{N_P} h_\ell P_\ell, \quad P_\ell \in \{I, X, Y, Z\}^{\otimes n_q}, \quad (62)$$

where

$$h_\ell = 2^{-n_q} \text{tr}(P_\ell \mathbf{H}). \quad (63)$$

A real symmetric $\mathbf{H}$ has zero coefficient on every Pauli string containing an odd number of Y factors, by the parity argument used in the proof of Proposition 4, so at most $2^{2n_q-1} + 2^{n_q-1} \leq 2N^2$ of the 4^{n_q} strings can be nonzero. The worst-case Pauli basis therefore still grows quadratically in the encoded structural dimension, at half the naive rate. Several consistent-mass cases below saturate this bound exactly—136 at $n_q = 4$ and 2080 at $n_q = 6$ —so it is attained, not merely an upper limit. Actual nonzero count depends on ordering, padding, mass model, and whitening. Commuting-group methods can reduce circuit settings but not eliminate finite-sampling error [29]. Throughout, a coefficient is counted as nonzero when $|h_\ell| > 10^{-10}$. That absolute cut is not innocuous for the consistent-mass operator, whose coefficient magnitudes show no numerical-zero plateau: at $N = 128$ the count is 2316, 2950, 3640 and 4251 for cuts 10^{-8} , 10^{-10} , 10^{-12} and 10^{-14} , and the fitted exponent of Section VIII-D moves from 1.36 to 1.65 over the same range. Lumped-mass counts are insensitive to the cut. Reported Pauli counts should therefore be read together with the threshold that produced them.

For a diagonal mass matrix the quadratic bound is far from attained, and the exact count is available in closed form.

Proposition 4 (Lumped-mass Pauli count). *Let the mesh be a d -dimensional grid with axis lengths $L_1, \dots, L_d$, each a power of two, indexed row-major so that axis a carries stride $\sigma_a = \prod_{b>a} L_b$, with nearest-neighbour couplings along each axis, and let $\mathbf{M}$ be diagonal so that $\mathbf{A} = \mathbf{M}^{-1/2} \mathbf{K} \mathbf{M}^{-1/2}$ inherits the sparsity pattern of $\mathbf{K}$. Then*

$$N_P \leq (n_q + 2) 2^{n_q-1} = O(N \log N), \quad (64)$$

independently of the spatial dimension d , with equality if and only if

$$S(x, z) = \sum_i (-1)^{|i \wedge z|} A_{i, i \oplus x} \neq 0 \quad (65)$$

for each of the n_q nonzero coupling masks x and every z with $|x \wedge z|$ even.

Proof. Write a Pauli string as $P = i^{n_Y} X^x Z^z$ with $n_Y = |x \wedge z|$. Then P connects only the index pairs $(i, i \oplus x)$, so a mask x

contributes nothing unless $A_{i, i \oplus x} \neq 0$ for some i . Along axis a the neighbour pairs are $(i, i + \sigma_a)$, and for a power-of-two axis length L_a the values $i \oplus (i + \sigma_a)$ realise exactly $\log_2 L_a$ distinct masks; the strides occupy disjoint bit ranges, so masks from different axes are distinct. The nonzero off-diagonal masks therefore number $\sum_a \log_2 L_a = \log_2 N = n_q$, with $x = 0$ contributing the diagonal.

Fix x and let $S(x, z) = \sum_i (-1)^{|i \wedge z|} A_{i, i \oplus x}$, so that the coefficient is $2^{-n_q} i^{n_Y} S$. Re-indexing $i \mapsto i \oplus x$ and using $\mathbf{A} = \mathbf{A}^T$ gives $S = (-1)^{|x \wedge z|} S$, so S vanishes whenever n_Y is odd. For $x \neq 0$ exactly half of the 2^{n_q} masks z satisfy $|x \wedge z|$ even, leaving 2^{n_q-1} surviving coefficients per nonzero mask, while $x = 0$ admits all 2^{n_q} . Hence N_P equals the number of pairs (x, z) with $|x \wedge z|$ even and $S(x, z) \neq 0$, which is at most $2^{n_q} + n_q 2^{n_q-1}$, with equality exactly under equation (65). $\square$

Condition equation (65) is checkable and not vacuous. For fixed x , $S(x, \cdot)$ is the Walsh–Hadamard transform of the vector $v_i^{(x)} = A_{i, i \oplus x}$ carried on that coupling diagonal, so equality asks only that no transform coefficient vanish on the even-parity indices. Heterogeneous properties make this generic; constant ones destroy it. On a homogeneous mesh $v^{(x)}$ is constant across its support, its transform concentrates on a vanishing fraction of z , and a uniform lumped chain carries only $3 \cdot 2^{n_q-1} - 1$ nonzero terms (5, 11, 23, 47, 95 for $n_q = 2, \dots, 6$), i.e. $\Theta(N)$ rather than $\Theta(N \log N)$. Evaluating equation (65) directly reproduces both regimes exactly: 20, 48 and 112 surviving pairs for the heterogeneous chains at $n_q = 3, 4, 5$, against 11, 23 and 47 for their uniform counterparts. The heterogeneous meshes used throughout this paper are chosen precisely to avoid such cancellations, so equation (64) is attained there while a uniform benchmark would report fewer terms. The count is also ordering-dependent: it characterizes the lumped operator in a locality-preserving indexing, and a scrambled DOF order moves it back toward the $O(N^2)$ worst case.

Equation (64) matches every lumped-mass count reported below exactly, in one, two, and three dimensions: 8, 20, 48, 112, 256, 576 for $n_q = 2, \dots, 7$, and 1280 and 6144 at $n_q = 8$ and 10 (Tables 2 and 5). It also explains why those counts coincide across spatial dimensions at matched N : the number of distinct coupling masks is $\log_2 N$ however the axes are partitioned. The measurement burden of the lumped model is therefore at most $(n_q + 2) 2^{n_q-1} = O(N \log N)$, not $O(N^2)$, and is attained by the heterogeneous meshes studied here; the quadratic regime is reached only through the mass model, which is the mechanism quantified in Section X.

Arbitrary amplitude-state preparation may itself require gate counts exponential in qubit number in the worst case, despite logarithmic qubit storage [30]. Resource reports must therefore include state preparation, two-qubit gates, circuit depth, Pauli terms, commuting groups, shots, and optimizer evaluations.

B. FINITE-SHOT PROPAGATION

For an independently measured Pauli mean μ_ℓ with n_ℓ shots,

$$\text{Var}(\hat{\mu}_\ell) = \frac{1 - \mu_\ell^2}{n_\ell}. \quad (66)$$

Ignoring covariance between groups,

$$\text{Var}(\hat{E}) = \sum_\ell h_\ell^2 \frac{1 - \mu_\ell^2}{n_\ell}. \quad (67)$$

For fixed total shots, the continuous optimal allocation satisfies

$$n_\ell \propto |h_\ell| \sqrt{1 - \mu_\ell^2}. \quad (68)$$

The delta method gives

$$\text{Var}(\hat{f}_i) \approx \left(\frac{s_t}{8\pi^2 f_i} \right)^2 \text{Var}(\hat{\varepsilon}_i). \quad (69)$$

The $1/f_i^2$ factor makes the fundamental frequency especially vulnerable when its eigenvalue is small compared with the total Pauli-estimator variance.

C. CONTROLLED ERROR BUDGET

An additive scalar “total error” is generally not identifiable because the components interact. We report the controlled error vector

$$\mathcal{E} = (\epsilon_{\text{FE}}, \epsilon_{\text{map}}, \epsilon_{\text{ansatz}}, \epsilon_{\text{opt}}, \epsilon_{\text{shot}}, \epsilon_{\text{channel}}, \epsilon_{\text{readout}}). \quad (70)$$

In this paper, mapping error is zero up to floating-point arithmetic; ansatz and optimizer error are separated using exact state vectors and multiple restarts; shot error is evaluated by Monte Carlo sampling; and a stylized channel/readout sensitivity is reported only as a diagnostic. Finite-element modelling error and real-device error remain unquantified and are explicitly excluded from claims.

No dimensionally mixed “information efficiency” is defined. Instead, the raw resource vector is reported:

$$\mathcal{R} = (n_q, N_P, N_G, N_{\text{shot}}, N_{\text{eval}}, D_{\text{circ}}, N_{2q}, C_{\text{asm}}, C_{\text{chol}}, C_{\text{bound}}, C_{\text{Pauli}}, C_{\text{prep}}). \quad (71)$$

where N_G is the number of measurement groups; C_{asm} , C_{chol} , C_{bound} , and C_{Pauli} denote classical matrix assembly, whitening, spectral-bound, and Pauli-construction costs; and C_{prep} is quantum state-preparation cost. When the global Gershgorin policy equation (24) is used, C_{bound} contains only the row-sum calculation. When a Ritz/Lanczos bound is used, its iterations, matrix–vector products, storage, and projected eigensolve are reported explicitly.

D. END-TO-END ASYMPTOTIC COST

The resource vector equation (71) collapses into asymptotic order for the tested one-dimensional chains, where $\mathbf{M}$ and $\mathbf{K}$ are banded with constant half-bandwidth b . In the following, k is the number of requested modes, N_P the number of nonzero Pauli terms (62), N_{eval} the number of optimizer evaluations, T_{it} the classical Lanczos iteration count, and ε the target accuracy of a single energy estimate. Table 1 compares a

TABLE 1. End-to-end asymptotic time and space for the k lowest modes of an N -DOF banded 1D chain (constant half-bandwidth b). Dashes mark steps the classical solver does not perform.[‡]

Stage	Classical (Lanczos)	Quantum (VQD)
Assembly	$O(N)$	$O(N)$
Cholesky of $\mathbf{M}$	$O(Nb^2)$ [¶]	$O(Nb^2) = O(N)$
Whitening $\mathbf{A}$	—	$O(N^2)$
Pauli decomposition	—	$O(N^2 \log N)$
State preparation [§]	—	$\Omega(N/\log N)$ depth
Solve / measure	$O(kN T_{\text{it}})$	$O(kN^2 \varepsilon^{-2} N_{\text{eval}})$
Total time	$O(kN T_{\text{it}})$	$O(kN^2 \varepsilon^{-2} N_{\text{eval}})$
Space	$O(Nk)$	$O(N) + O(N^2)$ [†]

T_{it} is the Lanczos iteration count: roughly constant with shift-invert ARPACK, $O(\sqrt{N})$ – $O(N)$ unpreconditioned. [†] $O(N)$ statevector amplitudes plus $O(N^2)$ storage for the whitened operator and its Pauli list. [¶]A generalized Lanczos also factorizes $\mathbf{M}$; negligible for banded $\mathbf{M}$. [§]Every circuit execution must first prepare the trial state, and Remark 3 bounds that depth below by $\Omega(N/\log_2 N)$. The quantum column therefore counts circuit executions and the classical column floating-point operations; the two are commensurable only after multiplying by that depth factor. [‡]Padding and the Gershgorin bound cost $O(N)$ and $O(N^2)$, both dominated by whitening. The tighter Ritz policy (25) instead costs $O(p \text{ nnz}(\mathbf{A}) + Np^2 + p^3)$ time and $O(Np)$ space, reported separately as C_{bound} . The whitening and Pauli rows assume band-aware triangular solves and a fast Walsh–Hadamard projection; the reference implementation uses dense LAPACK and an explicit term-by-term projection, so its wall time is not the tabulated asymptote.

sparse classical eigensolver with the variational quantum pipeline for the k lowest modes of an N -DOF chain.

Three consequences follow. First, preprocessing is not the bottleneck for these chains: the banded Cholesky is $O(N)$, not the $O(N^3)$ dense cost noted in Section III. Second, the classical solver acts on the sparse $\mathbf{K}$ directly, so its cost scales with $\text{nnz}(\mathbf{K}) = O(N)$, whereas whitening densifies $\mathbf{A}$ —from density 0.023 to 0.308 between lumped and consistent mass at $N = 128$ —so the Pauli representation carries $N_P = O(N^2)$ terms in the worst case ($2^{2n_q-1} + 2^{n_q-1} \leq 2N^2$). This quadratic figure is an upper bound: over the tested range the consistent-mass density and Pauli count grow sub-quadratically (empirical exponent ≈ 1.4 – 1.5 fitted over $N \geq 16$, Fig. 5), and no universal quadratic law is claimed. The local exponent moreover falls with N over $N \geq 8$, reaching 1.25 for the last tested doubling, so the growth is decelerating rather than settling onto a power law. Even so, merely constructing and storing that operator is superlinear in both time and space, and already exceeds the entire classical eigensolve once N is large relative to k , before a single shot is taken. Third, the measurement phase $O(kN^2 \varepsilon^{-2} N_{\text{eval}})$ dominates the quantum column; by equation (69) the low-frequency modes are the most sampling-intensive because the eigenvalue-to-frequency map amplifies variance as $1/f_i^2$. That expression treats ε as an absolute energy tolerance, which understates the growth: the decision-relevant requirement is a fixed relative accuracy on a frequency. With $\mathbf{H}$ normalized so $\|\mathbf{H}\| \approx 1$, $\text{Var}(\hat{E})$ is $O(1/n)$ independently of N , whereas $\lambda_1/\lambda_{\text{max}} = O(N^{-2})$ for a chain; holding the fundamental-frequency error fixed therefore demands $\varepsilon = O(N^{-2})$, hence $\varepsilon^{-2} = O(N^4)$ and a measurement term of $O(kN^6 N_{\text{eval}})$.

The mechanism is already visible at $N = 6$ in Table 4: the mode-1 to mode-4 RMSE ratio is 31.9 at 10^5 shots per term, against $(f_4/f_1)^2 = 31.6$. The classical cost is left as $O(kN T_{it})$ rather than a single power of N because T_{it} depends on the solver and the spectral gap; under either policy it is far below the quantum column. The optimizer factor N_{eval} and the shots needed per gradient enter only through trainability, and Section X finds the gradient variance decaying polynomially rather than exponentially in n_q at fixed shallow depth, i.e. a polylogarithmic factor in N —though that favourable reading does not survive increasing the depth toward expressibility. The no-advantage conclusion rests on the measurement bottleneck either way, and so does not depend on assuming an untrainable barren plateau.

IX. NUMERICAL PROGRAM

All experiments are reproducible with the accompanying Python code and use no quantum software development kit. Exact circuit states are constructed directly, Pauli decompositions are computed explicitly, and finite-shot estimates are sampled from the exact Pauli means. The studies are designed to answer separate questions rather than to make a speedup claim.

Two six-DOF benchmarks are used and should not be confused. Benchmark S is a fixed-base shear building with floor masses decreasing linearly from 220 to 170 t and story stiffnesses from 220 to 170 MN/m, giving $f_1 = 1.280$ Hz; it is scaled by $s_t = U_G$ and padded with $\alpha = 1.25$, and it backs the restart study of Section X and the certified-penalty run of Appendix B. Benchmark C is the $N = 6$ member of the fixed-base chain family of Test B, with $k = 2.2 \times 10^8(1 - 0.35\xi)$ N/m and element mass $2 \times 10^4(1 - 0.20\xi)$ kg for $\xi \in [0, 1]$ and lumped mass, giving $f_1 = 4.481$ Hz; it is scaled by $s_t = \lambda_{\text{max}}$ and padded with $\alpha = \max(1.10, U_G/s_t + 10^{-3})$, and it backs Tests A, D and E and the measurement floor. Both are one-dimensional and lumped-mass; the consistent-mass results are confined to Test B.

A. TEST A: EXACT DEFLATION THRESHOLD

Benchmark C is mass whitened and padded to three qubits. For targets $r = 2, 3, 4$, a common penalty is swept across the exact threshold in equation (22). Exact eigenvalues are used here only to verify the necessary-and-sufficient threshold; they are not counted as an operational preprocessing assumption. Thus $U_r = \varepsilon_r$ and $\delta_j = 0$ in the threshold verification. In a deployable execution, Algorithm 1 instead uses either the Gershgorin policy equation (24) or a recorded Ritz/Lanczos bound equation (25). A second experiment rotates each prior projector by a controlled angle and compares the target-state rotation with equation (30).

B. TEST B: HAMILTONIAN AND PREPROCESSING SCALING

Fixed-base chains with $N \in \{4, 8, 16, 32, 64, 128\}$ use smoothly varying stiffness and mass. Lumped and consistent

element mass matrices are compared. We record s_G/λ_{max} , transformed density, nonzero Pauli terms, and greedy qubit-wise commuting groups. Cholesky and transformation structure are profiled through $N = 512$. Wall-clock times are machine-specific and are not used as universal complexity claims.

C. TEST C: ANSATZ DEPTH, OPTIMIZER RESTARTS, AND TRAINABILITY

Real-amplitude circuits use a CNOT ring. Fundamental-mode VQE is run for 8 and 16 DOFs at depths 1–3. The six-DOF first mode is repeated for 20 random initializations at depths 1 and 2 to distinguish systematic expressibility failure from local optimizer failure. Gradient variance is sampled at 2–12 qubits under uniform and near-identity initialization.

D. TEST D: FINITE-SHOT MODAL FREQUENCY

For the exact first four modes of the six-DOF chain Hamiltonian, each Pauli term is measured independently with 10^2 , 10^3 , 10^4 , and 10^5 shots. Monte Carlo estimates are converted to frequency using equation (18). This isolates measurement propagation from ansatz and optimizer error.

E. TEST E: FINITE-SHOT DAMAGE LOCALIZATION

Stiffness losses of 2%, 5%, 10%, and 20% are introduced at element 3, counting from the base. The location is fixed rather than swept; across the six possible locations the exact normalized localization margin at 10% loss ranges from 0.515 to 0.702, so element 3 at 0.654 is mid-range and not favourably selected. Baseline and damaged element energy fractions are estimated independently from Pauli measurements. Both estimates use the baseline element operators $\mathbf{H}_e$ of equation (59); only the state and the normalization change between the healthy and damaged epochs. The element with maximum D_e is selected. Each condition uses 200 Monte Carlo trials; random guessing has probability $1/6 = 16.7\%$.

X. RESULTS

A. THE THEORETICAL PENALTY THRESHOLD IS SHARP

Fig. 2 shows that target-state overlap switches from zero to one at $\beta/\beta_{\text{min}} = 1$, up to the equality case where the ground state is degenerate. This numerical result verifies the theorem and replaces arbitrary tuning with a checkable spectral threshold.

Projector perturbations confirm the second half of the theory. The deflated separation is $g_r = 0.05$, so equation (30) applies while $\|\Delta_r\|_2 < g_r/2 = 0.025$. Within that range, projector rotation angles of 0.005 to 0.035 rad give perturbations $\|\Delta_r\|_2 = 0.0034$ to 0.0239 and target-state sine errors of 0.068 to 0.380, against Davis–Kahan bounds of 0.136 to 0.954. The bound holds at every point and is uniformly conservative by a factor of about two, correctly capturing the amplification of projector error by β_j . Beyond $\|\Delta_r\|_2 = g_r/2$ the guarantee lapses and the bound is vacuous: at a rotation of 0.06 rad the measured sine error is 0.508, but the target eigenvector is no longer certified to be isolated.

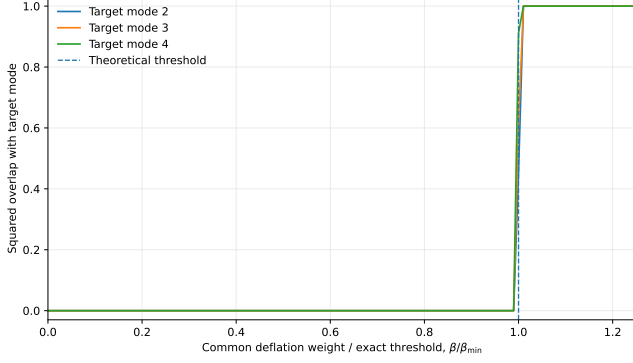


FIGURE 2. Exact deflation sweep for target modes 2–4. The target becomes the ground state only after the penalty exceeds the spectral threshold in equation (22).

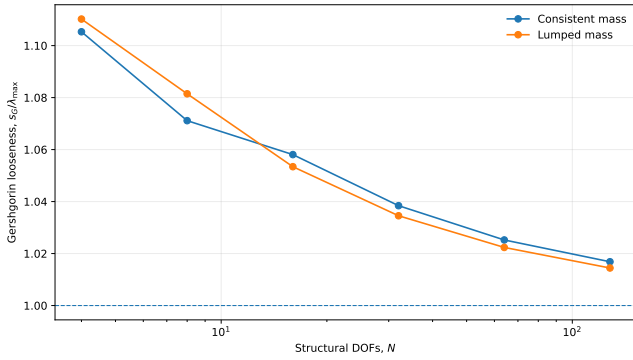


FIGURE 3. Gershgorin looseness for the structural-chain scaling study. The ratio approaches one with increasing N for these particular models.

The perturbation is deliberately adversarial. The first projector is rotated toward the target mode itself, so the contamination lies exactly along the direction deflation is meant to protect; this is why a 0.02 rad projector error produces a 0.249 target rotation, a twelvefold amplification.

B. GERSHGORIN SCALING IS NOT THE DOMINANT PROBLEM IN THE TESTED CHAINS

The empirical Gershgorin ratio is shown in Fig. 3. Across both mass models and $N = 4$ –128, the ratio lies between 1.014 and 1.110. Thus the bound is mildly conservative for these one-dimensional chains, although no universal tightness is claimed.

The two-scale scheme remains useful because a different topology or material contrast can produce a looser bound, and because certified padding should not depend on the optimizer normalization.

C. MASS WHITENING CREATES A MEASURABLE PAULI BOTTLENECK

Figs. 4 and 5 show the principal scaling result. At $N = 128$, lumped mass yields transformed density 0.0233 and 576 nonzero Pauli terms; consistent mass yields density 0.3077 and 2950 terms. At $N = 64$, greedy qubit-wise commuting

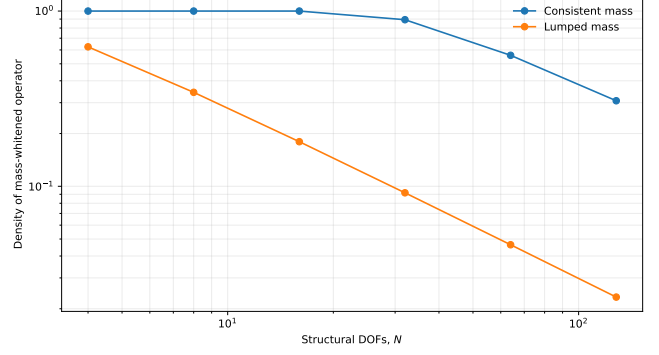


FIGURE 4. Density of the mass-whitened operator. Consistent mass introduces substantially more long-range coupling than lumped mass in the tested chains.

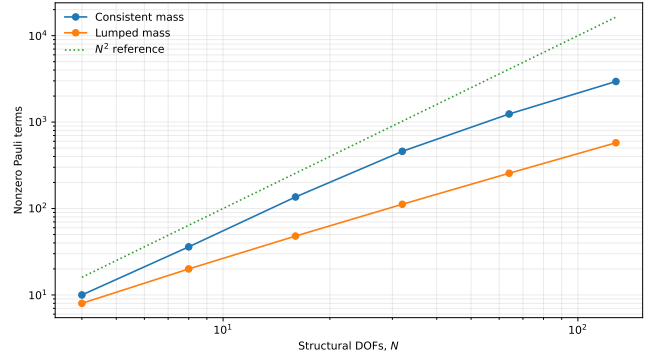


FIGURE 5. Nonzero Pauli-term count versus structural DOFs. The N^2 line is a reference, not a fitted asymptotic law.

groups increase from 64 for lumped mass to 221 for consistent mass.

Preprocessing to $N = 512$ confirms that the Cholesky factor of the chain remains banded, but $\mathbf{A}$ contains 1534 retained entries for lumped mass and 21554 for consistent mass. This is not evidence of universal dense scaling; it is direct evidence that whitening and mass formulation must be included in resource estimates.

D. CIRCUIT DEPTH, NOT QUBIT COUNT, CONTROLS EXACT-STATE RECOVERY

The 8- and 16-DOF fundamental-mode results are summarized in Table 3 and Fig. 6. For 16 DOFs, depth 1 gives 81.75% frequency error, depth 2 gives 5.16%, and depth 3 gives $2.27 \times 10^{-9}\%$. The exact-state result shows that four qubits are enough to store the vector but not that a shallow circuit can prepare it.

Twenty six-DOF restarts reinforce this interpretation. Depth 1 has a minimum first-mode error of 127.99%, a median of 130.20%, and a maximum of 130.20%; depth 2 has a median error $4.1 \times 10^{-12}\%$. The shallow failure is therefore systematic ansatz error, not an unlucky optimizer seed.

Remark 3 (Parameter-count lower bound on required depth). *The depth dependence in Table 3 follows from a parameter count. A real, normalized state in $D = 2^n$ dimensions has*

TABLE 2. Selected Hamiltonian-scaling results. QWC denotes greedy qubit-wise commuting groups.

Mass	N	n_q	$U_G/\lambda_{\max}$	Pauli terms	QWC groups
Lumped	16	4	1.053	48	16
Lumped	64	6	1.022	256	64
Lumped	128	7	1.014	576	–
Consistent	16	4	1.058	136	41
Consistent	64	6	1.025	1241	221
Consistent	128	7	1.017	2950	–

TABLE 3. State-vector VQE depth study for the fundamental mode. The failing rows are limited by ansatz expressibility and are stable under additional restarts; the near-zero entries are limited instead by optimizer tolerance and fall further as the restart budget is increased.

N	Qubits	Depth	Parameters	Frequency error (%)
8	3	1	6	27.33
8	3	2	9	2.00×10^{-12}
8	3	3	12	8.59×10^{-13}
16	4	1	8	81.75
16	4	2	12	5.16
16	4	3	16	2.27×10^{-9}

$D - 1$ free amplitudes, whereas the real-amplitude circuit of equation (38) at depth L has $n_q(L + 1)$ parameters. Reaching an arbitrary target therefore requires

$$n_q(L+1) \geq 2^{n_q} - 1, \quad \text{equivalently} \quad L \geq \frac{2^{n_q} - 1}{n_q} - 1. \quad (72)$$

A parameter count bounds the dimension of the reachable manifold, so it establishes that a fixed-depth family cannot be surjective onto the state sphere; it does not by itself prove that one particular target is unreachable. The condition is nonetheless consistent with every row of Table 3: for $n_q = 3$ it holds at $L = 2$ ($9 \geq 7$) but not $L = 1$ ($6 < 7$); for $n_q = 4$ it holds at $L = 3$ ($16 \geq 15$) but not $L = 2$ ($12 < 15$). Since $2^{n_q} = N$, the required depth grows as $L = \Omega(N/\log_2 N)$, so a fixed-depth family cannot remain expressive as N increases and the logarithmic qubit count does not translate into a logarithmic-depth circuit. The condition is necessary but not sufficient: the CNOT-ring entangler can further restrict the reachable manifold, so equation (72) is a lower bound on the depth an exact preparation requires. This is consistent with the systematic shallow-circuit failure reported above.

E. GRADIENT STATISTICS RAISE, BUT DO NOT SETTLE, THE BARREN-PLATEAU QUESTION

Two features of this diagnostic bound what it can establish, and both work against detecting a plateau rather than for it. The depth is held at $L = 2$ while n_q grows, so by Remark 3 the circuit is far from expressive over most of the range and comes nowhere near a unitary 2-design; the concentration mechanism that produces barren plateaus is therefore not something this experiment could exhibit even if it were operating. And the derivative is taken with respect to a single first-layer parameter, adjacent to $|0\rangle^{\otimes n_q}$, which is systematically the least plateau-prone choice available. The result below should be read as

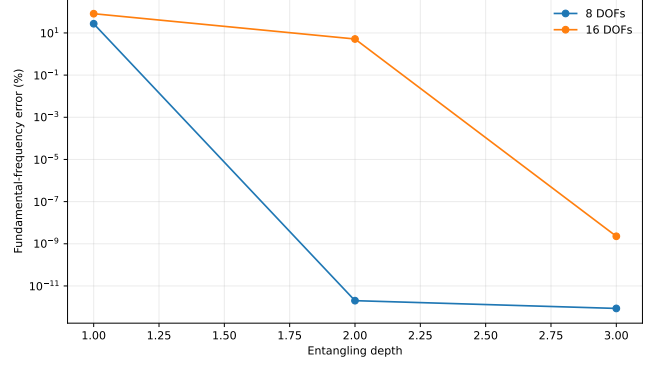


FIGURE 6. Fundamental-frequency error versus circuit depth for 8- and 16-DOF chain models.

evidence that no exponential decay has set in for this particular shallow family and this particular parameter, not as a probe of the asymptotic regime.

Fig. 7 reports gradient variance for a depth-2 circuit over 2–12 qubits, with bootstrap 95% confidence intervals from the 80 samples per point. Under uniform initialization the variance declines from 2.62×10^{-2} at 2 qubits to 9.82×10^{-4} at 12 qubits. A power law in n_q fits the uniform-initialization series better than an exponential: over the full range $R^2 = 0.96$ versus 0.83, favouring the power law by $\Delta\text{AIC} = 15$; over the small-system-free range $n_q \geq 4$ the advantage narrows to $R^2 = 0.90$ versus 0.85 ($\Delta\text{AIC} = 4$, moderate). The decline does not stop: a linear fit to the tail ($n_q \geq 7$) gives a shallow slope of -0.12 per qubit (95% CI $[-0.23, -0.01]$), statistically nonzero but far below the per-qubit exponential rate that defines a barren plateau. Near-identity initialization produces smaller and nonmonotonic values, including 1.52×10^{-5} at 12 qubits.

The decline is best read against the scale of the objective itself. Over the same range the variance of the energy across random parameters falls from 4.94×10^{-2} to 2.32×10^{-3} , so the normalized quantity $\text{Var}[\partial_a E]/\text{Var}[E]$ is statistically flat: a linear fit of its logarithm gives a slope of -0.036 per qubit (95% confidence interval $[-0.087, +0.016]$, $R^2 = 0.17$). Relative to the spread of the landscape it explores, the gradient does not shrink at all over $2 \leq n_q \leq 12$, and most of the raw decline in Fig. 7 reflects the concentration of the normalized spectrum rather than a loss of trainability. Thus at this fixed shallow depth the gradients shrink at an approximately polynomial, not exponential, rate.

That conclusion does not survive an increase in depth, and the fixed-depth diagnostic could not have revealed it. Repeating the sweep across $L \in \{2, 4, 8\}$, with the differentiated parameter drawn uniformly at random so the statistic is parameter-averaged rather than tied to θ_0 , the exponential decay rate steepens systematically: fitting $\log V_g$ against n_q over $2 \leq n_q \leq 10$ gives -0.42 per qubit at $L = 2$, -0.62 at $L = 4$ and -0.73 at $L = 8$, against the $-\ln 2 \approx -0.69$ that characterizes a barren plateau. Restricting the fit to $n_q \geq 7$ leaves the $L = 8$ figure at -0.78 . The $L = 8$ variance relative

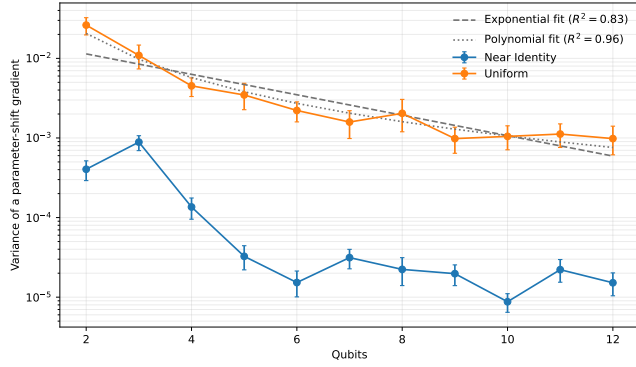


FIGURE 7. Empirical gradient variance for depth-2 real-amplitude circuits over 2–12 qubits, with bootstrap 95% confidence intervals and polynomial and exponential fits to the uniform-initialization series. The variance is better fit by a power law than an exponential; the finite range is a diagnostic, not an asymptotic barren-plateau proof.

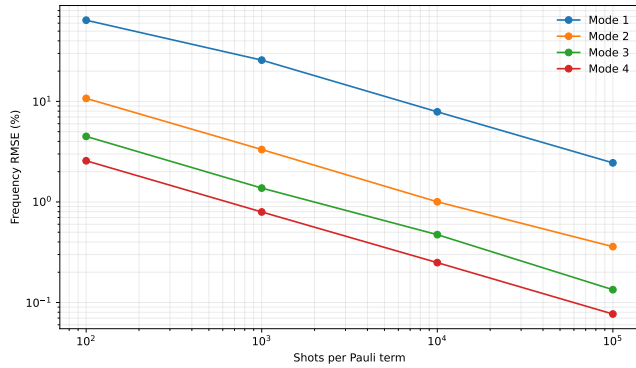


FIGURE 8. Finite-shot frequency RMSE from independent Pauli measurements of exact modal states.

to $L = 2$ falls monotonically for $n_q \geq 5$, from 0.83 to 0.12, so this is a depth–width interaction and not an artefact of differentiating one of more parameters.

The implication is unfavourable. By Remark 3 expressibility requires $L = \Omega(N/\log_2 N)$ —at $n_q = 10$ that is $L \geq 101$, so even $L = 8$ lies far inside the inexpressive regime—yet the barren-plateau rate is already attained there. Expressibility and trainability therefore pull against each other for this ansatz family: the depths that could represent the target are deeper than the depths at which gradient concentration has already set in. This is not a proof of an asymptotic plateau at $n_q \leq 10$, but it removes any reading in which the shallow-circuit result licenses optimism about deeper ones.

F. FINITE-SHOT FREQUENCY ESTIMATION IS MOST DIFFICULT FOR THE FIRST MODE

Fig. 8 and Table 4 show the independent-Pauli Monte Carlo results. At 10^4 shots per Pauli term, frequency RMSE is 7.88% for mode 1 but only 0.25% for mode 4. Even at 10^5 shots per term, mode-1 RMSE is 2.45%. This follows from equation (69): the square-root transformation amplifies eigenvalue uncertainty at small f_i .

The independent experiment is a baseline, not an optimized

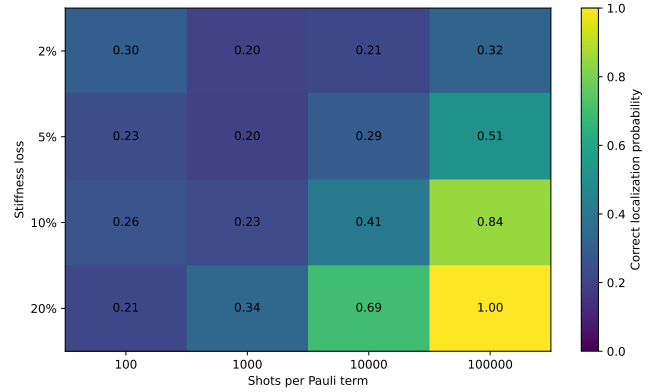


FIGURE 9. Correct damage-localization probability from independently sampled element-energy observables. The random-guess baseline is $1/6 = 16.7\%$.

measurement strategy. At matched total state preparations, exact-covariance QWC grouping lowers the theoretical RMSE floor by factors of approximately 2.5–2.8, but the first-mode floor remains 0.83%. Thus grouping is valuable yet does not remove the low-frequency sensitivity. Classical shadows, hardware-efficient entangled measurements, and error mitigation [31] should be compared in future work.

G. FINITE-SHOT DAMAGE LOCALIZATION IS ESTIMABLE, BUT ONLY GIVEN THE DAMAGED MODEL

Exact-state agreement of the modal energy fractions with the classical solution is a consistency check only. The finite-shot localization study is more informative. Fig. 9 shows that at 10^5 shots per Pauli term a 10% stiffness loss is localized correctly in 84.5% of 200 trials (Wilson 95% interval approximately 78.8–88.9%), against a 16.7% random baseline, and a 20% loss in every trial. The median localization margin is positive at this budget (+0.28 for 10% loss), meaning the damaged element typically outscores every intact competitor.

The budget dependence is steep. The same 10% loss is localized in only 40.5% of trials at 10^4 shots per term (33.9–47.4%) and remains only marginally above guessing at 10^3 (23.0%, one-sided binomial $p = 0.013$ against $1/6$). Small damage remains hard at every budget tested: a 2% loss reaches only 32.0% at 10^5 shots per term, with a still-negative median margin. Localization therefore succeeds in this benchmark, but only for moderate damage and only at measurement budgets that Section XII quantifies as the dominant end-to-end cost.

Remark 4 (Sensitivity to the element-operator convention). *The choice in equation (58) to build the numerator from the baseline $\mathbf{K}_e^0$ is consequential, not cosmetic. If the operator is instead built from the damaged element stiffness $\mathbf{K}_e(\mathbf{d})$ —an oracle that presumes the quantity being inferred—the explicit $(1 - d_e)$ factor opposes the denominator and mode-shape changes and cancels most of the signal: the exact localization margin at 10% loss falls by a factor of 11.2, measured accuracy drops from 84.5% to 25.0%, and the certified shot requirement rises by two orders of magnitude. Estimating a*

TABLE 4. Frequency RMSE (%) under independent Pauli sampling and an optimistic QWC lower bound.

Measurement budget	Mode 1	Mode 2	Mode 3	Mode 4
10^2 shots/term, independent	64.20	10.72	4.49	2.57
10^3 shots/term, independent	25.73	3.32	1.37	0.79
10^4 shots/term, independent	7.88	1.00	0.47	0.25
10^5 shots/term, independent	2.45	0.36	0.13	0.08
QWC-optimal floor at matched total budget [†]	0.83	0.14	0.054	0.032

[†]The six-DOF Hamiltonian has 19 nonidentity Pauli terms and eight greedy qubit-wise-commuting groups. The independent rows are 400-trial Monte Carlo estimates, carrying a relative standard error of about 3.5% (so the mode-1 entry is good to roughly ± 0.09); the floor is analytic, and the improvement factors quoted in the text compare it against the *analytic* independent RMSE (2.285% for mode 1), not against the Monte Carlo row above, so dividing the tabulated rows directly gives a slightly different ratio. The floor uses the same 1.9×10^6 total shots per mode as the 10^5 -shots-per-term baseline, exact within-group covariances, continuous optimal allocation, perfect simultaneous measurement, and no circuit, readout, or allocation overhead. It is therefore an optimistic lower bound, not an expected hardware result.

difference of operator expectations is therefore sensitive to which operator difference is taken, a point that applies to quantum change-detection observables generally.

A cross-check against the frequency results of Section X sharpens the picture. The same 10% loss shifts the modal frequencies by -1.20% , -0.16% , -1.75% and -0.24% for modes 1–4, while the corresponding finite-shot RMSEs at 10^5 shots per Pauli term are 2.45%, 0.36%, 0.13% and 0.08%. The damage signal is thus below the measurement noise for the two lowest modes and an order of magnitude above it for the third. Frequency-based detection of this damage is possible at this budget, but only through the higher modes; the fundamental, on which most vibration-based monitoring practice rests, is the least usable of the four.

The study tests the damage formulation end to end and identifies measurement variance, not the formulation, as the limiting factor. Direct independent-Pauli estimation remains inefficient for small stiffness changes; grouped or shadow-based measurement of the change observable is the obvious next step.

XI. GENERALIZATION TO TWO AND THREE DIMENSIONS

Every experiment above uses a one-dimensional chain, the most favorable case for Cholesky fill. This section tests whether the mapping and deflation theory carry over unchanged to higher spatial dimension, and whether the densification finding of Sections III and VIII is a property of the 1D chain specifically or worsens as dimension increases.

A. WHAT REMAINS DIMENSION-AGNOSTIC

Nothing in the theory assumes a specific spatial dimension. The Cholesky mapping and mass–Euclidean isometry (Proposition 1) hold for any $\mathbf{M} \succ 0$, $\mathbf{K} = \mathbf{K}^T$. Spectral-safe padding (Proposition 2) and the scalar-scaling law (Proposition 3) depend only on the spectrum of $\mathbf{A}$. The exact deflation theorem and the Davis–Kahan perturbation bound (Theorem 1, equation (30)) hold for any Hermitian operator with a spectral gap, independent of what physical system it was built from. The parameter-count depth bound (Remark 3, equation (72)) depends only on n_q , not on how the DOFs are arranged in space. The resource vector equation (71) and the finite-

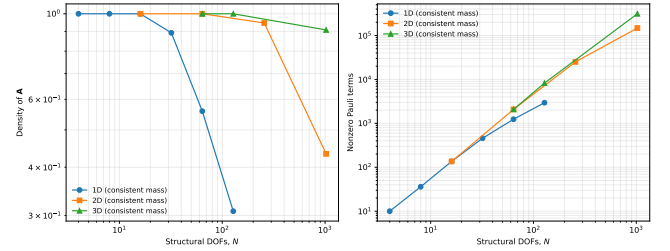


FIGURE 10. Transformed-operator density and nonzero Pauli-term count for consistent mass, 1D chains versus the 2D and 3D grids of Table 5. The density-decline onset shifts to larger N as dimension increases: 1D declines by $N \approx 32$ –64, 2D only marginally by $N = 256$ and clearly by $N = 1024$, and 3D remains 91% dense at the largest tested size.

shot propagation formulas (equations (66), (67) and (69)) are likewise generic to any Pauli-decomposed Hamiltonian. Only the values taken by N_P , the transformed density, and the Gershgorin ratio are specific to a given mesh, and those are what this section measures.

B. TWO- AND THREE-DIMENSIONAL SCALING

The mesh is a scalar grid of point masses connected by nearest-neighbor springs, the direct d -dimensional generalization of the 1D chain: the same two-node element pattern is applied along d axes instead of one, with one DOF per node so that the comparison isolates spatial dimension rather than DOFs per node. Free-DOF counts are fixed at exact powers of two by fixing an entire boundary layer along one axis, generalizing the chain’s fixed base. Stiffness and mass vary smoothly and asymmetrically across the grid axes, exactly as the 1D study varies properties along the chain, so that no accidental grid symmetry inflates or deflates the sparsity count. Table 5 and Fig. 10 report density and nonzero Pauli terms for lumped and consistent mass at matched qubit counts.

Lumped mass remains markedly sparser than consistent mass at every tested size in every dimension, as in the 1D case. Densities of exactly 1.000 in Table 5 are rounded: the 2D $N = 64$ and 3D $N = 128$ operators carry 2079 and 8251 nonzero Pauli terms against the real-symmetric maxima 2080 and 8256, the small deficits being accidental cancellations rather than structural sparsity. For consistent mass, the infor-

TABLE 5. Two- and three-dimensional scalar-mesh scaling, generalizing Table 2. Grid sizes are chosen as exact powers of two, as in the 1D study.

Dimension	Mass	N	n_q	Density of $\mathbf{A}$	Pauli terms
2D	Lumped	16	4	0.250	48
2D	Lumped	64	6	0.070	256
2D	Lumped	256	8	0.019	1280
2D	Lumped	1024	10	0.005	6144
2D	Consistent	16	4	1.000	136
2D	Consistent	64	6	1.000	2079
2D	Consistent	256	8	0.947	25082
2D	Consistent	1024	10	0.433	147620
3D	Lumped	64	6	0.086	256
3D	Lumped	128	7	0.045	576
3D	Lumped	1024	10	0.006	6144
3D	Consistent	64	6	1.000	2080
3D	Consistent	128	7	1.000	8251
3D	Consistent	1024	10	0.908	312325

mative comparison is at matched N : at $N = 1024$, 2D density has fallen to 0.433 while 3D remains at 0.908—the same DOF count, a visibly different regime. This is consistent with nested-dissection theory: 2D separators have size $O(\sqrt{N})$ and 3D $O(N^{2/3})$, giving fill $O(N \log N)$ and $O(N^{4/3})$ respectively [15], so the onset of sparsity-recovering behavior is expected to shift to larger N as dimension increases, and 3D should lag furthest behind. Fitting a single power law for the nonzero Pauli count over the full tested range gives an increasing but modest exponent (1D ≈ 1.66 , 2D ≈ 1.69 , 3D ≈ 1.79 ; the 3D fit rests on only three sizes). These are full-range fits: restricting the 1D fit to $N \geq 16$ gives 1.48 instead, which is the figure quoted in Section VIII-D, and the 1D local exponent continues to fall with N . Because density has not yet stabilized into a clean power-law regime in any dimension over the tested sizes, these exponents are reported as illustrative fits, not asymptotic laws, exactly as Section X already cautions for the 1D case. The Gershgorin ratio remains mild in every case tested (1.04–1.14), consistent with the 1D finding of Fig. 3 that Gershgorin looseness is not the dominant issue.

Declining density is a fractional effect: the absolute nonzero Pauli-term count continues to grow in every dimension tested (Table 5), so this does not indicate that the measurement burden diminishes with N . Density and N_P are structural properties of the operator, not complexities in themselves; they separately drive the space row and the measurement term of the time row in Table 1. The practical relevance of eventual sparsity recovery is also limited by scale. For an idealized cubic mesh, the graph diameter (maximum node-to-node hop distance) is $d(N^{1/d} - 1)$ in d dimensions; matching the diameter of the $N = 128$ 1D chain already tested (127 hops) would require an idealized square mesh of $N \approx 4,160$ (12 qubits) or an idealized cubic mesh of $N \approx 81,000$ (16.3 qubits). This is an illustrative order-of-magnitude estimate for a regular cubic mesh, not a claim about arbitrary geometries, but it places any such recovery for 3D structures far beyond near-term hardware and, likely, beyond the regime in which classical simulation of the whitened operator itself remains convenient.

These results do not extend to irregular topology, block

DOFs, or realistic 3D frames; they show that the 1D densification finding is not an artifact of one spatial dimension, and that it does not improve—if anything it worsens and appears later—as dimension increases.

C. VALIDATING THE DEGENERACY DIAGNOSTICS

A 1D chain with smoothly varying properties has no repeated eigenvalues and so can never exercise the subspace-based diagnostic of equation (47). A symmetric two-dimensional grid can: reflecting the grid across its diagonal is an exact symmetry, so a uniform square grid has genuinely degenerate mode pairs. Using a uniform (not the heterogeneous) grid from Section XI-B, an adjacent-eigenvalue pair was found with relative gap 2.2×10^{-16} , i.e. degenerate to floating-point precision.

For a degenerate pair, any rotation of the two eigenvectors within their shared subspace is an equally valid choice, so two independent solves (or a solver and a slightly perturbed rerun) may return differently oriented, but physically identical, bases. Let $\phi_a(\theta) = \cos \theta \phi_a + \sin \theta \phi_b$ be the pair rotated by angle θ within the degenerate subspace. Comparing ϕ_a to $\phi_a(\theta)$ via equation (45) gives, since both are M-orthonormal,

$$\text{MAC}_{aa}^{(M)}(\theta) = \cos^2 \theta, \quad (73)$$

exactly—confirmed numerically to machine precision. Fig. 11 shows this degrading from 1 to 0.5 at $\theta = 45^\circ$ to essentially 0 at $\theta = 90^\circ$, which by conventional MAC thresholds would be flagged as a severe mismatch, even though the physical subspace never changed. The subspace-angle diagnostic equation (47) is stated in the whitened representation, where the isometry of Proposition 1 makes a plain transpose product equivalent to an M-weighted one; written directly in the physical representation, it evaluates ε_{sub} between $\text{span}(\phi_a, \phi_b)$ and its rotated copy via $(\mathbf{Y}_C)^T \mathbf{M} \mathbf{Y}_Q$. Since the rotation is an isometry of the shared subspace, this overlap matrix is exactly a 2×2 rotation matrix, whose singular values are exactly one, giving $\varepsilon_{\text{sub}} = 0$ for every θ . The measured values stay at the eigensolver’s numerical floor ($\sim 3 \times 10^{-8}$) across the full sweep. Since $\varepsilon_{\text{sub}} = 0$ follows analytically, this confirms the implementation rather than discovering the

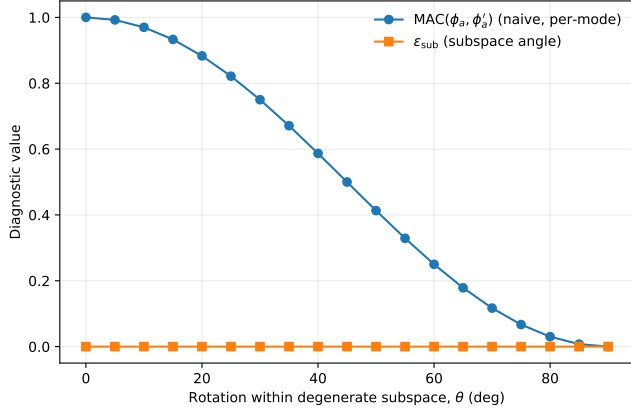


FIGURE 11. Naive per-mode MAC versus the subspace-angle diagnostic for a genuine degenerate pair, rotated within its own subspace. MAC degrades exactly as $\cos^2 \theta$ despite no physical change; ϵ_{sub} correctly remains at the numerical floor throughout.

behaviour; the informative case, a near-degenerate pair under a physical perturbation where ϵ_{sub} is small but nonzero, is not exercised here.

This is a direct demonstration of why Section VII’s subspace comparison exists: a per-mode diagnostic can manufacture a false structural-health alarm from an arbitrary, physically meaningless basis choice whenever a structure has near-symmetric geometry, and the subspace-angle diagnostic is immune to exactly that failure mode by construction.

XII. DISCUSSION

A. WHAT IS NOW THEORETICALLY GUARANTEED

Three properties have exact statements. First, Cholesky whitening preserves eigenvalues and converts mass orthogonality to Euclidean orthogonality. Second, safe padding places artificial states above a certified physical bound for any positive optimization scale. Third, exact deflation identifies mode r if and only if every lower-mode penalty exceeds its spectral difference from the target. With approximate prior states, equations (28) and (30) quantify the resulting loss of accuracy.

No theorem is claimed for global convergence of the parameterized circuit optimizer. This distinction is essential: a correct deflated operator can still be inaccessible to a shallow ansatz or missed by a nonconvex optimizer.

B. WHY HEURISTIC COMPOSITE PENALTIES ARE UNSAFE

A weighted cost $E + \gamma\sigma_H^2 + \zeta\ell_p$ changes the optimization landscape without a universal conversion between energy, residual, and leakage. An adaptive multiplicative β rule is uncontrolled in both directions: it can stall just above the spectral threshold, where equation (30) is vacuous, or grow without reference to G_r , where the certificate degrades linearly. Temporal overlap and frequency penalties can suppress genuine damage-induced changes. A scalar exponential integrity score can conceal a severe failure in one component behind good performance in

another. Finally, a mixed-dimension “information efficiency” is uninterpretable without a reference task and cost model.

The proposed framework therefore uses theorem-based penalties, constrained diagnostics, independent epoch solutions, non-aggregated validation vectors, and raw resource counts.

C. THE ACTUAL SCALING BOTTLENECK

For the tested chain models, Gershgorin looseness is modest. The more serious issue is that consistent-mass whitening changes operator structure, increasing nonzero Pauli terms and measurement groups; Section XI shows this bottleneck does not ease, and its onset shifts to larger N , as spatial dimension increases. The density of $\mathbf{A}$ is not an independent design variable: for fixed $\mathbf{K}$ and DOF ordering it is a deterministic consequence of the mass model through the Cholesky factor. The design tension is therefore a two-term causal chain rather than a trilemma,

$$\begin{array}{c} \text{physical fidelity of } \mathbf{M} \\ \xrightarrow{\text{Cholesky}} \text{density of } \mathbf{A} \\ \xrightarrow{\text{Pauli}} \text{measurement cost.} \end{array} \quad (74)$$

in which only the physical fidelity of $\mathbf{M}$ and the resulting measurement cost are genuinely in opposition; the intermediate density of $\mathbf{A}$ is induced, and is further modulated by DOF ordering and coefficient thresholding. Lumped mass may be cheaper quantum mechanically but can perturb high modes; consistent mass may be more faithful but measurement-heavy. A credible study must report both modal discretization error and quantum resource consequences. For the chains studied here the discretization penalty is small precisely where the quantum method operates. Taking consistent mass as the reference, the lumped model reproduces the first four frequencies to within 3.88% at $N = 16$, 0.25% at $N = 64$, and 0.06% at $N = 128$; the disagreement grows with mode index, reaching 4.9% at mode $N/4$ and 42% at the highest mode. Since the finite-shot RMSE of the fundamental is 2.45% at 10^5 shots per Pauli term (Table 4), the extra fidelity bought by consistent mass at $N = 128$ is roughly forty times smaller than the measurement noise it must be read through, while costing 2950 Pauli terms against 576. For low-mode targets at achievable shot budgets the expensive mass model is therefore not merely optional but unusable—an argument for the lumped operator and its $O(N \log N)$ term count of Proposition 4, not for the consistent one.

As an illustration, consider the $N = 128$, $k = 4$ consistent-mass chain of Table 2. Table 6 contrasts the two routes with order-of-magnitude figures.

The gap spans several orders of magnitude and is dominated by measurement, not preprocessing, which supports the qualitative statement that classical methods are overwhelmingly preferable for the present benchmarks. Any regime in which a quantum route could become competitive lies outside the variational setting studied here: it would require either fault-tolerant quantum phase estimation rather than sampling-based

TABLE 6. Illustrative end-to-end comparison for the $N = 128$, $k = 4$ consistent-mass chain. Preprocessing is finite-element assembly (classical) versus Cholesky factorization, whitening, and Pauli decomposition (quantum). Quantum figures are order-of-magnitude estimates, not measured device results.

Quantity	Classical	Quantum
Qubits	—	7
Preprocessing	$O(N)$	$O(N^2 \log N)$
Nonzero Pauli terms	—	2950
Core work	~ 1400 matvecs	$\sim 10^{11}$ circuits [‡]
Wall time	< 1 s	$\sim 3.5 \times 10^5$ s

[‡] $\sim 10^5$ shots per term $\times$ 2950 terms $\times$ a few hundred optimizer evaluations $\times$ 4 modes; the wall time assumes an optimistic 1μ s per circuit execution with no queuing, calibration, or readout overhead.

VQE, or operators that are genuinely dense so that classical sparsity offers no advantage—neither of which applies to sparse structural finite-element models. This is stated only to delimit scope and is not a claim of attainable advantage.

D. IMPLICATIONS FOR STRUCTURAL HEALTH MONITORING

The finite-shot damage result changes the interpretation of quantum modal strain energy. The expectation-value identity is mathematically valid and practically reachable, but only at a measurement cost that must be stated explicitly. Detection requires resolving a small difference between two noisy ratios of expectation values. For baseline and damaged estimates $\hat{a}_d/\hat{b}_d$ and $\hat{a}_0/\hat{b}_0$, a first-order variance is

$$\text{Var}(\hat{\eta}_d - \hat{\eta}_0) \approx \mathbf{g}_d^T \Sigma_d \mathbf{g}_d + \mathbf{g}_0^T \Sigma_0 \mathbf{g}_0, \quad (75)$$

$$\mathbf{g} = [1/b \quad -a/b^2]^T.$$

For equal shots n per nonidentity Pauli term, write $\text{Var}(\hat{D}_e) \approx V_{D,e}/n$. A normal-approximation design criterion for detecting a nonzero score at two-sided type-I error α and power $1 - \beta$ is

$$n_{\text{det}} \gtrsim \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 V_{D,e^*}}{D_{e^*}^2}. \quad (76)$$

For localization, let $e^{(2)}$ be the strongest intact competitor and $\Delta_D = D_{e^*} - D_{e^{(2)}}$. A conservative Bonferroni pairwise requirement is

$$n_{\text{loc}} \gtrsim \frac{(z_{1-\alpha/(n_e-1)} + z_{1-\beta})^2 V_{\Delta_D}}{\Delta_D^2}, \quad (77)$$

where V_{Δ_D} is the per-shot variance coefficient of the score margin. The absolute-value signs are linearized at their exact signs. These expressions are design approximations, not finite-sample guarantees. They are also matched to the alternative rather than to the null: D_e is a sum of absolute values, so under $D_e = 0$ the estimator is folded and biased upward by $\sigma\sqrt{2/\pi}$ per term, and the nominal type-I error is not attained. A deployed test should calibrate its null by permutation rather than from equation (76).

The calculation uses the first four modes with equal weights, independent baseline/damaged numerator and denominator estimates, and independent element-score variances. It is conservative because beneficial shared-denominator covariance is ignored. For 10% damage, 10^5 shots per term give a localization signal-to-noise ratio of 1.44, consistent with the observed 84.5% success. Approximately 8.7×10^4 shots per term are required to detect the damaged-element score with 80% power, whereas family-wise localization against all five intact competitors requires roughly 4.8×10^5 per term. With 648 nonidentity operator-term settings across four modes, six elements, and two structural states, the latter corresponds to approximately 3.1×10^8 circuit shots under the independent baseline—four modal states measured to part-in- 10^3 precision for a single binary maintenance decision. That figure counts every (operator, term) pair as a distinct setting and is therefore an upper bound. Each element operator's Pauli support is contained in the 19 nonidentity strings of $\tilde{\mathbf{A}}$ itself, exactly so for the chains at $N \leq 16$ and to within 12% of the union at $N = 32$, and $\sum_e \mathbf{H}_e = \tilde{\mathbf{A}}$; one measurement of those 19 strings per mode and structural state therefore serves all seven operators. Counting distinct settings instead gives $8 \times 19 = 152$ and $\approx 7.3 \times 10^7$ shots. Re-using the records introduces precisely the numerator–denominator covariance that equation (75) discards, so the two figures bracket the achievable cost. Measurement therefore remains the dominant end-to-end cost even where the decision itself is reachable, which makes covariance-aware joint measurement and direct estimation of change observables the most promising levers on that cost.

E. END-TO-END QUANTUM ADVANTAGE REMAINS UNESTABLISHED

Any future advantage claim must include:

- assembly and reduction of $\mathbf{M}$ and $\mathbf{K}$;
- factorization or generalized-quotient preparation;
- Hamiltonian construction and Pauli decomposition;
- state preparation and repeated measurements;
- optimizer evaluations and higher-mode deflation;
- readout of decision-relevant structural observables;
- comparison with sparse classical eigensolvers at matched accuracy.

For the present benchmarks, classical methods are overwhelmingly preferable.

XIII. LIMITATIONS AND RESEARCH AGENDA

The main structural models remain one-dimensional chains, the most favorable case for Cholesky fill; Section XI reports an initial two- and three-dimensional scalar-mesh generalization confirming that densification does not improve, and appears later in N , as dimension increases. That study nonetheless uses regular scalar grids, not the block DOFs, irregular topology, constraints, and material heterogeneity of real bridges and buildings, and the measured exponents must not be extrapolated to such structures or to alternative DOF orderings and block-sparse formulations. Pauli decomposition is evaluated

TABLE 7. Delta-method shots per Pauli term for the six-DOF damage example ($\alpha = 0.05$, 80% power). In equations (76) and (77) only, β denotes the type-II error rate, not a deflation penalty; the detection criterion is two-sided and the localization criterion one-sided with a Bonferroni correction over the five intact competitors.

Stiffness loss	Detect damaged-element change	Localize against all five competitors
2%	2.55×10^6	1.52×10^7
5%	3.85×10^5	2.23×10^6
10%	8.75×10^4	4.83×10^5
20%	1.82×10^4	9.01×10^4

only under binary amplitude encoding. The finite-shot study uses independent Pauli measurements and exact input states; it does not include circuit noise, state-preparation error, grouped covariance, or device calibration. The damage study uses stiffness-only damage and no environmental variability; long-term monitoring campaigns show that environmental effects can rival or exceed damage-induced modal changes [32], [33]. It also supplies the damaged mode shapes as exact eigenvectors of a known damaged operator, so it isolates measurement variance and does not model the identification problem: in deployment the damaged states would be obtained from measured response data, not from an eigensolve of the very model being inferred, whereas the element operators $\mathbf{H}_e$ of equation (59) are correctly built from the baseline model alone. Cholesky timings are platform-specific, while the structural sparsity counts are more portable.

The next high-impact study should combine:

- (i) a laboratory frame or bridge dataset with FDD/SSI uncertainty;
- (ii) block-sparse three-dimensional finite elements and alternative DOF orderings;
- (iii) Cholesky, generalized quotient, and domain-decomposed mappings;
- (iv) grouped or shadow-based measurement of modal observables;
- (v) near-degenerate subspace tracking under temperature and damage;
- (vi) a matched end-to-end comparison against ARPACK/Lanczos and reduced-order classical models.

XIV. CONCLUSIONS

This paper establishes a stricter evaluation standard for variational quantum modal analysis. The mass-orthonormal Cholesky map is physically consistent, but its preprocessing and densification costs are not negligible. Gershgorin bounds can certify padding without fixing the optimizer scale. Higher-mode penalties admit an exact threshold, and approximate-projector perturbation theory shows why oversized penalties are dangerous. Variance and leakage should be diagnostics or constrained refinements, not arbitrarily weighted additions to the primary energy. Temporal mode identity should be assigned after independent eigensolution so that continuity cannot mask damage.

The expanded experiments show both promise and difficulty. Increasing circuit depth recovers small exact-state modal problems, while a shallow ansatz fails systematically despite sufficient qubit dimension. In the tested chains,

consistent mass substantially increases Hamiltonian density and Pauli count, and an initial two- and three-dimensional generalization shows this effect worsening, not improving, with spatial dimension, while also validating a subspace diagnostic on a genuine degenerate pair unavailable in one dimension. Finite-shot frequency and damage estimates remain costly—localizing a 10% stiffness loss with certified family-wise confidence needs $\sim 10^8$ circuit executions—with the lowest mode particularly sensitive to sampling variance. These results do not establish quantum advantage. They identify the mathematical conditions and resource bottlenecks that a future structural quantum algorithm must overcome.

The main conclusion is therefore precise: qubits are necessary to encode the state, but spectrally valid modes, expressive circuits, sparse observables, and decision-level measurement precision are necessary to solve the engineering problem.

DATA AND CODE AVAILABILITY

The accompanying archive contains the complete LaTeX source, bibliography, generated figures, CSV results, and Python scripts required to reproduce the exact-state, scaling, trainability, finite-shot, and damage-localization studies, together with the two- and three-dimensional generalization and degeneracy-diagnostic study of Section XI.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

APPENDIX A REPRESENTATIVE FINITE-ELEMENT MATRICES

These are quoted for reference only. Every model solved in this paper is a scalar spring-mass chain or grid with one degree of freedom per node; no truss or beam element is used in any experiment. For a two-dimensional truss element with $c = \cos \theta$ and $s = \sin \theta$,

$$\mathbf{K}_e = \frac{EA}{L} \begin{bmatrix} c^2 & cs & -c^2 & -cs \\ cs & s^2 & -cs & -s^2 \\ -c^2 & -cs & c^2 & cs \\ -cs & -s^2 & cs & s^2 \end{bmatrix}. \quad (78)$$

For an Euler-Bernoulli beam bending element,

$$\mathbf{K}_e = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}, \quad (79)$$

Algorithm 1 Mass-orthonormal sequential VQE with provable deflation thresholds

Require: Reduced $M \succ 0$, $K = K^T$; target modes m ; ansatz family; tolerances.

- 1: Compute $M = LL^T$ using a sparse ordering where appropriate.
- 2: Form $A = L^{-1}KL^{-T}$ by triangular solves.
- 3: Compute certified U_G and choose optimization scale s_t .
- 4: Form H_p with $\alpha = U_G/s_t + \delta_p$.
- 5: Decompose the required operators and group compatible measurements.
- 6: **for** $r = 1, \dots, m$ **do**
- 7: Select and record the bound policy: $U_r = U_G/s_t$, or compute $U_r = \vartheta_r^{(p)} + \delta_{fp}$ by a p -dimensional Ritz/Lanczos pass and record C_{bound} .
- 8: **for** $j < r$ **do**
- 9: Obtain δ_j from exact validation data or the declared residual-plus-shot confidence rule in equation (26).
- 10: Set $\beta_j = U_r - \hat{\varepsilon}_j + \delta_j + \tau$.
- 11: **end for**
- 12: Minimize $C_r(\theta) = \langle \psi(\theta) | \hat{H}_r | \psi(\theta) \rangle$ over multiple initializations.
- 13: Optionally solve the constrained variance refinement in equation (36).
- 14: Reject or flag the state if any component of equation (44) exceeds its declared tolerance.
- 15: Recover f_r and ϕ_r using equations (18) and (40).
- 16: **end for**

$$M_e = \frac{\rho AL}{420} \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}. \quad (80)$$

APPENDIX B SEQUENTIAL MODAL ALGORITHM WITH PROVABLE DEFLATION THRESHOLDS

For the reproducible six-DOF exact-state validation, the declared acceptance set was

$$\begin{aligned} \epsilon_f &\leq 10^{-4}\%, \quad \rho \leq 10^{-5}, \quad \sigma_H^2 \leq 10^{-10}, \\ \ell_p &\leq 10^{-8}, \quad o \leq 10^{-8}, \quad 1 - \text{MAC}^{(M)} \leq 10^{-6}. \end{aligned} \quad (81)$$

These values are numerical verification tolerances, not universal SHM thresholds; several are unattainable at any sampling budget, the variance bound $\sigma_H^2 \leq 10^{-10}$ most obviously so, and $\langle H_p^2 \rangle$ carries its own measurement cost that equation (71) does not itemize. They accept all four depth-2 exact-state modes and reject the depth-1 results. Field applications must replace them with tolerances calibrated to model discrepancy, environmental variability, and decision consequences.

Algorithm 1 was executed on this benchmark under the quantum-only Gershgorin policy equation (24), so that no auxiliary classical eigensolve enters the procedure. With $s_t = U_G$ the certified bound is $U_r = 1$, the radius δ_j is the residual $\sigma_{H,j}$ returned by the run itself, and $\tau = 10^{-3}$. At depth 2 the resulting penalties lie between 0.68 and 0.98:

they satisfy the threshold of Theorem 1 by construction rather than by tuning, and they adapt to solution quality as intended—at depth 1, where the extracted modes are poor and $\sigma_{H,j}$ is correspondingly large, the same rule returns 0.81 to 1.02, automatically widening the margin to absorb the worse projectors. They are also three to four and a half times smaller than the ad hoc common penalty $\beta = 3$ used for comparison. Measured against equation (32) rather than against the bare threshold, the Gershgorin policy places the excess $U_r - \varepsilon_r + \delta_j + \tau$ at 2.0, 2.9 and 4.5 times G_r for targets $r = 4, 3, 2$, i.e. always past saturation and never in the divergent regime, at the cost of a Davis–Kahan certificate loosened by factors of 1.3, 1.8 and 3.2 relative to the optimum. Conservatism in the bound policy is what buys this margin. At depth 2 the certified run recovers all four modes with frequency error below $10^{-12}\%$, mass-weighted MAC equal to one to twelve digits, prior-mode overlap below 4×10^{-15} , and padding leakage below 10^{-15} , meeting the acceptance set above with several orders of margin; at depth 1 it fails for the ansatz reasons of Remark 3, not because of the penalty rule.

APPENDIX C EXACT RELATION BETWEEN PROJECTOR ERROR AND STATE ANGLE

For normalized vectors u and $\hat{u}$, the rank-one projector difference satisfies

$$\|\hat{u}\hat{u}^T - uu^T\|_2 = \sin \angle(\hat{u}, u). \quad (82)$$

Therefore the deflation perturbation may be written as

$$\|\Delta_r\|_2 \leq \sum_{j < r} \beta_j \sin \angle(\hat{u}_j, u_j), \quad (83)$$

which gives a direct rule for balancing penalty margin against the quality of previously extracted modes.

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